SEN381 | SOFTWARE ENGINEERING 381

CivicConnect

Milestone 3 Project Engineering Document

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BASELINE STATUS

Team approved M3 engineering baseline
Conditionally ready for local demonstration

DOCUMENT VERSION AND DATE

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GitHub Repository

https://github.com/RynoGz/SEN831_Group_F

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0. Document control

This document continues the approved M1 and M2 engineering record as PED v3.0. Requirement identifiers and original acceptance criteria are retained, with FR-009 AC6 controlled through CR-M3-01. Sections 13 to 18 preserve the dated M2 snapshot. The current RTM, registers and M3 sections include final local and hosted verification, migration 008, the detail-read performance correction and the remaining work scheduled for the final submission. Final verification identifies the tested source through application commit 374c21e, successful CI13 and the source-hash manifest in section 32.

Version	Date	Purpose	Owner or approval
0.2	7 September 2026	Member 1 foundation draft	Ryno Goetz
0.3	8 September 2026	Requirements and RTM draft	Steven Riaan Piek
0.9	8 September 2026	Unified M1 review draft with corrected registers	All members
1.0	8 September 2026	Controlled M1 engineering baseline	Approved by Ryno Goetz, Steven Riaan Piek and Willem Booysen
1.1	30 September 2026	M2 contribution drafts, implementation and review evidence	All members
2.0	30 September 2026	Controlled M2 architecture, design and application baseline	Approved by Ryno Goetz, Steven Riaan Piek and Willem Booysen
3.0	16 October 2026	Integrated M3 implementation, local and hosted verification, defect corrections and final-submission follow-up; final CI/review reconciliation	Team-approved M3 contributions and CR-M3-01; Ryno authorised final verification and recorded M4 follow-up

0.1 Baseline approvals

M1 was approved on 8 September 2026 and M2 on 30 September 2026. The three M3 contributions were approved through PRs 30, 31 and 32 on 13 October. Ryno confirmed PED approval and enforced branch protection. All three members jointly selected and approved CR-M3-01 on 15 October. Steven and Willem approved its implementation, as confirmed by Ryno. The candidate and evidence are identified in section 32.

1. Problem and business need

1.1 Problem statement

CivicConnect addresses a loss of control over the lifecycle of service requests. The organisation receives operational requests through several informal and disconnected channels, but those channels do not provide one dependable record connecting a request to its owner, current state and actions taken. A request can therefore reach someone without becoming visible work that can be assigned, followed up and verified as completed [S4, section 2].

This fragmentation affects three groups differently. Requesters cannot reliably determine what happened after submission. Staff must reconcile information and ownership across channels while organising work. Management cannot confidently distinguish outstanding work from completed work or reconstruct why a request changed state. Inconsistent handling of sensitive request information compounds the visibility and accountability problem [S4, sections 2-2.1].

The underlying engineering problem is consequently more than collecting request descriptions: the platform must preserve useful information and controlled responsibility from submission through feedback and authorised closure. Improving the interface alone would not establish that control if assignment, permitted status changes, access boundaries and reporting definitions remained ambiguous. This is analysis derived from the scenario, not an observation from a conducted stakeholder interview.

1.2 Causes, effects and intended response

Request information is spread across messages, calls, spreadsheets and paper

Work may be missed or duplicated; reconciliation consumes staff effort.

Proposed engineering response: Maintain a retrievable digital request record with controlled classification and clear submission outcome. **Stakeholder/need link:** STK-001, STK-002; NEED-001, NEED-002.

Ownership and progress are difficult to see

Requesters repeatedly seek updates; staff may work on the same request or leave it unowned.

Proposed engineering response: Expose authorised status/history, explicit responsibility and lifecycle feedback. **Stakeholder/need link:** STK-001, STK-002, STK-004; NEED-003 to NEED-005.

Status changes and actions lack dependable accountability

A completion claim is difficult to explain or check.

Proposed engineering response: Control state changes and retain attributable action/resolution information. **Stakeholder/need link:** STK-002, STK-003; NEED-006, NEED-008.

Management reporting is manual and inconsistent

Workload and overdue work cannot be assessed consistently.

Proposed engineering response: Derive oversight views from request records using agreed definitions and useful category/status dimensions. **Stakeholder/need link:** STK-003; NEED-007, NEED-008.

Sensitive information is exchanged inconsistently

Visibility can expose details to people who do not need them.

Proposed engineering response: Define role-related access and sensitive-data handling early; later verify access with positive and negative tests. **Stakeholder/need link:** STK-001, STK-002, STK-005; NEED-009.

The scenario problems are grounded in the scenario [S4, section 2]. Consequences and responses are the team's proposed analysis. They do not imply that message import, automatic duplicate detection or a particular authentication technology is required.

1.3 Business need

The organisation needs an affordable, supportable platform that lets requesters submit and follow requests, lets authorised staff take responsibility and manage them, and gives management a dependable view of service activity. Its value depends on the same request information supporting operational work and oversight, with access and actions controlled throughout the lifecycle [S4, sections 2.1 and 3].

The proposed product scope therefore prioritises the required request-management workflow and sufficient accountability. Automatic imports from existing channels, payments and predictive routing are excluded or deferred as proposals because they would create additional integration, security, testing and cost obligations without being necessary to demonstrate the required minimum capabilities [S4, section 3.1; sections 3.4-3.5].

1.4 Intended value and how to assess it

The following evaluation methods turn business value into checkable outcomes. They are not measured implementation results. The requirements section translates them into requirements and acceptance criteria using the approved M1 working policies and measurable baseline targets [S4, sections 5, 11 and 15].

VAL-001

A successfully submitted request remains findable.

Proposed validation evidence: Submit an agreed set of valid requests and check that each acknowledged request can subsequently be retrieved by its authorised requester. **Scope/need link:** SCP-001, SCP-004; NEED-001, NEED-003.

VAL-002

Requesters can see meaningful progress.

Proposed validation evidence: Walk through accepted, rejected, updated and completed outcomes; verify that feedback and the visible state agree with the recorded outcome. **Scope/need link:** SCP-003, SCP-005; NEED-003.

VAL-003

Staff can identify responsibility and permitted next actions.

Proposed validation evidence: Exercise assignment/acceptance and authorised state changes; verify ownership, retained action information and rejection of unauthorised changes. **Scope/need link:** SCP-009 to SCP-012, SCP-017; NEED-005, NEED-006, NEED-009.

VAL-004
Management can rely on its work overview.

Proposed validation evidence: Use a known set of requests with agreed overdue rules and compare open/overdue/resolved/closed views and category/status breakdowns with expected records. **Scope/need link:** SCP-013 to SCP-016; NEED-007, NEED-008.

VAL-005
Better visibility does not expose restricted information.

Proposed validation evidence: Verify requester and staff visibility against the agreed permission matrix, including attempts to access another user's restricted request. **Scope/need link:** SCP-006, SCP-017; NEED-009.

VAL-006
The solution remains workable within team and operating constraints.

Proposed validation evidence: Review actual scope delivery, schedule variance, quality/security evidence, service limits and likely running cost at later milestones. **Scope/need link:** CON-001 to CON-010; NEED-010, NEED-011.

No percentage reduction in lost requests, turnaround time or operating cost is claimed. The brief provides no measured current-process baseline. Such claims would require a defined before/after comparison and trustworthy measurements.

1.5 Boundaries and related requirements

The initial category set is Maintenance, IT support, Facility fault, Damaged equipment, Security concern, Lost property and Other. A request records a title, description, category, location, requester identity/contact details, automatic submission date/time, a sensitive-information indicator and an optional attachment. The software does not itself repair equipment or provide physical emergency response. The team should validate this working taxonomy during requirements review and control later changes.

The requirements section uses the stakeholder needs and proposed success evidence as inputs. The risk and forward-engineering sections address the risks implied by lost ownership, inconsistent reporting and sensitive-data access. The team should judge success using stakeholder value, controlled scope, time, resources, quality, security and operational evidence, rather than a successful demonstration alone [S4, sections 5 and 21].

References: Source register, especially S4 sections 2-5, 11, 15 and 21; S1 section 3.

2. Stakeholder analysis

2.1 Method and evidence limits

This is a scenario-based analysis. Requester, staff and management roles come directly from the Master brief [S4, section 3]. Coordination, operational support and assessment roles below are analytical groupings inferred from the described responsibilities and lifecycle controls. They do not assert that separate departments or named client representatives exist. One person may perform several organisational roles; the final permission model must distinguish responsibilities rather than assume one job title equals one software role.

Influence means ability to affect scope, priorities, acceptance or policy. Interest means how directly the stakeholder is affected by the system's outcomes. High/medium ratings are qualitative judgements for engagement planning, not measured survey results. All ratings and conflicts require review.

2.2 Stakeholder register

STK-001

Requesters; explicit in S4 section 3.

Needs and expectations: Submit understandable, appropriately classified requests; see their status/history and meaningful feedback; protect sensitive details. **Influence and interest:** Medium / high: directly affected, but no policy authority is stated.

Proposed engagement: Validate submission and feedback scenarios; ask what acknowledgement and progress information are useful.

STK-002

Authorised service staff; explicit in S4 section 3.

Needs and expectations: Find relevant work and details, take responsibility, apply permitted state changes and record actions/resolutions. **Influence and interest:** High / high: workflow feasibility and data quality depend on staff use.

Proposed engagement: Walk through assignment, reassignment, rejection, resolution and closure cases; verify access and required information.

STK-003

Management/oversight; explicit in S4 section 3.

Needs and expectations: Dependable activity views, identification of overdue/open/resolved/closed work, accountability and useful analysis. **Influence and interest:** High / high: influences priorities, reporting definitions and acceptance.

Proposed engagement: Agree reporting meaning and visibility; validate summaries against a known request set.

STK-004

Service coordination responsibility; inferred from assignment/prioritisation problems in S4 section 2; may be held by STK-002 or STK-003.

Needs and expectations: Clear responsibility, controlled categories and consistent handling of unassigned or incorrectly routed work. **Influence and interest:** High / high for workflow policy.

Proposed engagement: Clarify who may assign, change category, reassign or close work; validate boundary cases.

STK-005

Organisation's operational/security support responsibility; inferred from S4 sections 16-18; named owner unconfirmed.

Needs and expectations: Appropriate access, manageable configuration, protected credentials, recoverability and sustainable operation. **Influence and interest:** Medium / high; actual authority must be confirmed.

Proposed engagement: Review sensitive-data questions and later deployment/support constraints without choosing a stack in M1.

STK-006

Project team: Ryno, Steven and Willem; S2 and S4 sections 7.1-8.

Needs and expectations: Clear scope and dependencies, feasible workload, shared understanding and maintainable engineering evidence. **Influence and interest:** High / high for engineering choices within the brief.

Proposed engagement: Daily WhatsApp/Discord discussions, meetings every second day after 11:00, issues, meaningful two-reviewer PRs and joint baseline review.

STK-007

SEN381 lecturer/assessor; S4 sections 19, 20 and 23.

Needs and expectations: A defensible product and authentic individual/team evidence that meets the project controls. **Influence and interest:** High / high for assessment; not assumed to be the real organisation's management.

Proposed engagement: Clarify ambiguous assessment requirements and demonstrate controlled evidence at milestones.

2.3 Stakeholder needs for requirements derivation

Need ID	Need	Stakeholder/source	Priority basis and scope link
NEED-001	A valid new request can be recorded with enough information to act on it	STK-001, STK-002; S4 section 3, requester submission	Required minimum; SCP-001
NEED-002	Requests use a controlled category mechanism that supports routing and reporting	STK-001, STK-004; S4 section 3, requester categorisation	Required minimum; SCP-002
NEED-003	Requesters can find their earlier requests, inspect current status and understand submission/lifecycle outcomes	STK-001; S4 section 3, requester visibility and feedback	Required minimum; SCP-003 to SCP-005

Need ID	Need	Stakeholder/source	Priority basis and scope link
NEED-004	Staff can locate relevant authorised work and inspect its details using useful search/filter/sort criteria	STK-002; S4 section 3, staff visibility/search/details	Required minimum; SCP-006 to SCP-008
NEED-005	A request has explicit responsibility that can be assigned or accepted under agreed permissions	STK-002, STK-004; S4 sections 2 and 3	Required minimum; SCP-009
NEED-006	Changes, actions and resolution/closure follow defined responsibilities and permitted transitions	STK-002, STK-004; S4 section 3, staff lifecycle and actions	Required minimum; SCP-010 to SCP-012
NEED-007	Management can identify service activity and distinguish open, overdue, resolved and closed work	STK-003; S4 section 3, management oversight	Required minimum; SCP-013, SCP-014
NEED-008	Management can analyse work by useful dimensions and obtain sufficient information to explain responsibility and progress	STK-003; S4 section 3, management analysis/accountability	Required minimum; SCP-015, SCP-016
NEED-009	Access and disclosure match justified responsibilities, including treatment of sensitive request information	STK-001, STK-002, STK-005; S4 sections 2, 3 and 16	Cross-cutting security obligation; SCP-006, SCP-017
NEED-010	The platform can be verified, operated and recovered using available resources	STK-005, STK-006; S4 sections 4, 15, 17 and 18	Lifecycle obligation; SCP-018; CON-002, CON-009, CON-012
NEED-011	The team can demonstrate controlled collaboration and defend evidence individually	STK-006, STK-007; S4 sections 8-11 and 19	Project/process obligation; CON-001, CON-007, CON-008

These needs are not the final functional/non-functional requirements. Steven must add unique requirement IDs, priority, precise behaviour and measurable acceptance criteria and link them through the RTM [S4, section 11].

2.4 Meaningful conflicts and proposed resolutions

Visibility versus confidentiality

Requesters and management want useful progress information; staff/support must prevent inappropriate disclosure.

Approved resolution: Separate requester-visible progress from restricted details and apply the least-privilege baseline. Consequence for later work: M2 must refine the permission matrix for UI/API access checks, data exposure rules and negative security tests; this does not justify an unrestricted management role.

Quick submission versus actionable information

Requesters benefit from a short form; staff need enough context to categorise and act.

Proposed resolution for validation: Agree only the necessary required fields, validate them clearly, and record how missing information is handled. **Consequence for later work:** Affects validation, acceptance criteria, data collection and usability; field list remains Q-001.

Flexible handling versus accountable status

Staff need to correct assignments and update work efficiently; management needs consistent history.

Proposed resolution for validation: Define authorised transitions and retain attributable changes; decide how correction, rejection and closure are represented. **Consequence for later work:** Shapes lifecycle requirements, reporting definitions and tests; exact transitions remain Q-003.

Rich reports versus sustainable scope

Management could benefit from exports and predictive analysis; the three-person team has limited capacity.

Proposed resolution for validation: Deliver the mandatory oversight views first; propose advanced analytics as future scope subject to impact analysis. **Consequence for later work:** Preserves required reporting while controlling extra security, implementation, testing and cost obligations.

Fast completion versus independent review

Ryno wants to finish promptly; team governance requires both other members to review substantive changes.

Proposed resolution for validation: Submit coherent artefact units early and reserve review time within daily capacity. **Consequence for later work:** Requires authentic PR/review evidence and prevents approval becoming a last-minute dependency.

These conflicts are reasoned from the scenario and project constraints, not claims of observed stakeholder disputes.

2.5 Requirements traceability examples

Example A - requester visibility. S4 section 3 -> STK-001 -> NEED-003 -> SCP-003/SCP-004. Candidate behaviour: an authorised requester can retrieve the current status and history/list of their submitted requests. Candidate acceptance scenario: seed two requesters with distinct requests; verify each sees the correct own list and current state after a permitted staff update. Whether either may see another person's requests must follow the agreed access policy, with denied-access cases included. Final FR/NFR and acceptance IDs are recorded in the requirements and RTM sections.

Example B - overdue reporting. S4 section 3 -> STK-003 -> NEED-007 -> SCP-014. Working behaviour: a request is overdue when its agreed target date/time has passed and it is not Resolved, Closed or Rejected. The target is set by authorised staff or a manager during acceptance/assignment. Candidate acceptance scenario: use records either side of the due-time boundary and records in excluded states; compare the result to this rule. Steven should convert the rule into a testable requirement and the team should formally approve it.

These examples demonstrate currently available traceability. Design, implementation, test execution and release evidence are added later, not marked complete now [S1, section 3; S4, section 11.1].

References: Source register, S4 sections 2-4, 8, 11, 16-19; S1 section 3; S2 and S3.

3. Scope baseline

3.1 Scope objective and system boundary

CivicConnect will support the digital management of service requests from submission and classification through authorised staff handling, feedback and management oversight. The boundary includes request records, responsibility, state changes, relevant action/resolution information, role-appropriate views and meaningful service reporting. Physical fulfilment of the requested work remains an organisational activity outside the software [S4, sections 2-3].

The initial product serves one organisation. The scope below carries every minimum capability in the Master brief into the approved M1 baseline and records the working policies needed for requirements engineering. It defines the agreed scope but does not claim completed implementation [S4, sections 3 and 11; S1, section 3].

3.1.1 Selected request and workflow policy

- Required request information: title, description, category, location, requester identity/contact, automatically recorded submission date/time and a sensitive-information indicator; an attachment is optional.
- Initial categories: Maintenance, IT support, Facility fault, Damaged equipment, Security concern, Lost property and Other.
- Lifecycle: Submitted -> Accepted -> Assigned -> In Progress -> Resolved -> Closed. Submitted requests may be Rejected; authorised users may reassign; a requester or manager may request reopening; every status change must be attributable.
- Overdue rule: the agreed target date/time has passed and the request is not Resolved, Closed or Rejected. Authorised staff or a manager sets the target on acceptance or assignment.
- Feedback: show in-application feedback for accepted, rejected, updated, resolved and closed requests. External delivery channels are deferred.

3.2 In-scope minimum business capabilities

All items SCP-001 to SCP-016 represent required minimum capabilities from S4 section 3. Their detailed behaviour, fields, policies and verification conditions are linked to Steven's requirements and acceptance criteria. "Required" is the priority basis from the brief; these items are baselined in PED v1.0.

Scope ID	Included capability	Stakeholder/need	Boundary and clarification
SCP-001	Submit a new service request with appropriate information	STK-001; NEED-001	Use the selected fields in section 3.1.1; no automatic import from existing channels is assumed
SCP-002	Categorise a request through a controlled mechanism	STK-001, STK-004; NEED-002	Use the initial category set in section 3.1.1; controlled changes remain possible; automated AI classification is not required
SCP-003	View the current status of submitted requests	STK-001; NEED-003	Apply the agreed visibility policy; make the status meaningful to requesters
SCP-004	View a history/list of previously submitted requests	STK-001; NEED-003	Preserve retrievability under the agreed retention policy; not a promise of indefinite storage

Scope ID	Included capability	Stakeholder/need	Boundary and clarification
SCP-005	Receive meaningful feedback on accepted, rejected, updated and completed requests	STK-001; NEED-003	Provide in-application feedback for accepted, rejected, updated, resolved and closed events; external channels are deferred
SCP-006	View work relevant to authorised staff	STK-002; NEED-004, NEED-009	Define relevant/authorised visibility; staff access must not be assumed universal
SCP-007	Search, filter or sort requests using useful criteria	STK-002; NEED-004	Agree useful criteria with staff; the brief does not prescribe an advanced search engine
SCP-008	View full request details when authorised	STK-002; NEED-004	Full details are permission-dependent, including sensitive information
SCP-009	Assign or accept responsibility for a request	STK-002, STK-004; NEED-005	Authorised staff may accept/assign work; a coordinator or manager may assign and reassign it
SCP-010	Update status through controlled transitions	STK-002, STK-004; NEED-006	Apply the lifecycle and attributable-change rule in section 3.1.1
SCP-011	Record relevant actions, comments or resolution information	STK-002; NEED-006	Define required accountability and requester-visible versus restricted information
SCP-012	Resolve or close requests where authorised	STK-002, STK-004; NEED-006	Clarify resolved versus closed, rejection and any reopening policy
SCP-013	View useful service activity information	STK-003; NEED-007	Agree an oversight view suited to the organisation's decisions
SCP-014	Identify open, overdue, resolved and closed work	STK-003; NEED-007	Apply the overdue definition in section 3.1.1 and verify its time boundary
SCP-015	View requests by category/status or other justified dimensions	STK-003; NEED-008	Core category/status visibility is included; additional dimensions require value justification
SCP-016	Access enough information for accountability and service-performance analysis	STK-003; NEED-008	Retain sufficient attributable request history; this does not require predictive analytics or a data warehouse

3.3 Cross-cutting obligations

Scope ID	Included obligation	Source and implication
SCP-017	Appropriate access and sensitive-data protection throughout the request lifecycle	S4 sections 2, 4 and 16; supports NEED-009. Define role/access and disclosure rules now; defer the authentication technology choice, not the security requirements
SCP-018	Measurable quality, verification and controlled delivery/operation across later milestones	S4 sections 15-18; supports NEED-010. M1 identifies concerns/requirements; design, tests, staging, release, recovery and operational evidence develop through later milestones

These obligations remain part of the overall project even though their full implementation is not an M1 deliverable.

The access baseline applies least privilege: requesters submit and view their own requests; authorised staff view relevant work and perform permitted updates; coordinators/managers assign, reassign and view reporting; management sees authorised summaries/details without blanket access to sensitive information; and administration rights do not

automatically grant business-data access. Use fictional, non-sensitive test data. The exact retention period will be confirmed later because neither supplied brief fixes one.

3.4 Recommended out-of-scope items for the initial product baseline

The following are recommended for team sign-off. They are not exclusions stated by the lecturer and must not remove a capability required above.

OUT-001

Live two-way ingestion/synchronisation with WhatsApp, email, telephone systems or paper workflows.

Justification: The existing channels explain fragmentation; S4 section 3 does not require integrations. Supporting several channels adds reconciliation, privacy, credentials, dependency and testing obligations. **Effect and revisit condition:** Requests are entered through CivicConnect under SCP-001. Revisit if validated stakeholder evidence makes integration necessary and the team approves its impact.

OUT-002

Billing, payments and procurement/inventory management.

Justification: They are separate business processes beyond the stated request lifecycle and add security and reconciliation work. **Effect and revisit condition:** Record/request operational work without implementing financial or stock-management systems. Revisit only through justified scope change.

OUT-003

Performing repairs, emergency dispatch or guaranteeing real-world service completion times.

Justification: The software records and coordinates work; physical fulfilment and emergency response require organisational resources beyond the software boundary. **Effect and revisit condition:** Security-related concerns may still be recorded. Validate user-facing escalation expectations; do not market the platform as an emergency-response service.

Defence of OUT-001: a messaging integration could reduce repeated entry, but it must also resolve identity, duplicate submissions, update conflicts, access and delivery failures. The required business value can first be demonstrated through controlled submission and tracking inside CivicConnect. Proposing this exclusion preserves all minimum capabilities while keeping work compatible with three members' capacity. Its trade-off is an adoption/manual-entry burden, which should be recorded as a risk for Willem to assess rather than ignored [S4, sections 3.1, 4, 12 and 18].

3.5 Recommended future or deliberately deferred work

FUT-001

Native mobile applications and offline synchronisation.

Why it is not a current commitment: These create additional platforms, synchronisation states and verification effort beyond the minimum capabilities. **Information required before deciding:** User/device/connectivity needs, accessibility evidence, schedule and maintenance impact.

FUT-002
AI routing, duplicate detection and predictive service analytics.

Why it is not a current commitment: Basic controlled categories and oversight can satisfy the minimum capabilities; automated inference adds accuracy and accountability questions. **Information required before deciding:** Evidence of value, suitable data, failure handling, privacy and operational-cost analysis.

FUT-003
Bulk migration of historic email, spreadsheets and paper records.

Why it is not a current commitment: Legacy sources are fragmented and their formats, quality and access permissions are not supplied. **Information required before deciding:** Sample formats, volume, reconciliation/retention policy and a justified migration requirement.

FUT-004
External message delivery such as SMS, WhatsApp or email notifications.

Why it is not a current commitment: Required feedback under SCP-005 remains included; external channels are not yet justified. **Information required before deciding:** Agreed feedback timing/channel needs, delivery failure handling, service limits, cost and sensitive-content policy.

FUT-005
Final technology stack, architecture and detailed persistence/UI/API design.

Why it is not a current commitment: These are later lifecycle decisions, explicitly outside required M1 decisions/implementation. **Information required before deciding:** Baselined FRs/NFRs, constraints, team capability, environment compatibility and comparison evidence.

In-application feedback is the selected M1 working policy for SCP-005. Deferring an external channel must never silently remove meaningful feedback. Likewise, FUT-003 does not remove required history for requests submitted to the new platform.

Willem should record only genuine team decisions or justified deferments in the Decision Log after review; these proposals are decision inputs, not approved entries [S4, section 13].

3.6 M1 milestone boundary

M1 is due on 9 September 2026 and produces the controlled problem, stakeholder, scope, requirements, constraints, traceability, risk, forward-thinking, decision and governance foundation in PED v1.0. Final stack/architecture selection, detailed database/UI/API or design-pattern implementation, CI pipeline implementation, extensive application code and production deployment are not required M1 deliverables unless approved as exploration [S1, sections 3-5; S3].

This separates when the team performs work from what the final product must support. Security, testability, deployment and recovery must be considered now even though implementation occurs later.

3.7 Approval and controlled change

Steven checked every SCP item and selected policy against the requirements and acceptance criteria and linked source and stakeholder needs through the RTM. Willem reviewed the risks, forward concerns and decisions. All three members approved the exclusions, controlled deferrals and unified PED v1.0 baseline [S4, section 11 and Appendix D].

After approval, changes require a recorded request and impact analysis, a decision and authorisation, controlled implementation and verification, then updates to the affected baseline artefacts. Analyse scope, schedule/resources, cost, quality/security, requirements, design/data/interfaces, tests, deployment and risks as applicable; use S4 Appendix E rather than silently absorbing additions [S4, section 14].

References: Source register, S4 sections 2-4, 11-18 and Appendices D-E; S1 sections 3-5.

4. Constraints and assumptions

4.1 Constraint register

The Master brief defines constraints on scope, four-milestone delivery, cost, quality, security and technology decisions. The following analysis applies those constraints to this team and CivicConnect's request workflow [S4, sections 4 and 18]. Proposed IDs support traceability and do not replace Willem's Risk Register.

CON-001

Exactly three team members; each must contribute and defend the full project.

Source/status: S4 sections 7.1-8; S2 records Ryno, Steven and Willem. **Engineering implication and planned response:** Assign primary owners and both non-author reviewers. Plan review capacity; specialisation does not remove shared understanding.

CON-002

All three members are available every day for four hours each.

Source/status: S3; confirmed by Ryno. **Engineering implication and planned response:** Gross daily capacity is 12 person-hours. Discussion, meetings, review and corrections count within it; record actual effort separately.

CON-003

Delivery through four formal milestones; M1 is due 9 September 2026.

Source/status: S4 sections 4 and 20; S3 confirms the M1 date. **Engineering implication and planned response:** Complete author checks and handoff early enough for two genuine non-author reviews; later milestone dates still need the formal schedule.

CON-004

WhatsApp/Discord, daily discussions and a meeting every second day after 11:00.

Source/status: S3; working arrangement agreed by all three members. **Engineering implication and planned response:** Use frequent updates for dependencies; transfer material decisions/actions to controlled repository records.

CON-005

Every minimum business capability must remain in controlled product scope; additions need justification.

Source/status: S4 sections 3-4; SCP-001 to SCP-016 map the minimums. **Engineering implication and planned response:** Protect the core lifecycle and oversight. Assess integrations/analytics separately; do not silently drop feedback, overdue views or authorised closure.

CON-006

Requirements and quality expectations need traceable, measurable evidence.

Source/status: S4 sections 11 and 15; S1 section 3. **Engineering implication and planned response:** Define useful performance/usability/reliability criteria and test conditions with Steven. Avoid unsupported adjectives or invented traffic targets.

CON-007
One controlled repository, protected main, PRs and two independent non-author approvals; no substantive direct development on main.

Source/status: S4 section 9; all three members have full repository access, but protection/evidence must be checked. **Engineering implication and planned response:** Use task branches and meaningful reviews for documents as well as code. A one-approval setting alone would not meet the brief's two-approval rule.

CON-008
PED/RTM evolve with controlled history; evidence and AI accountability must be authentic.

Source/status: S4 sections 6, 8, 10-11 and 23. **Engineering implication and planned response:** Preserve identifiers/version history, record real contributions and verification, and integrate approved Member 1 content into the PED.

CON-009
Prefer free/low-cost services where practical while recognising limits and likely ongoing costs.

Source/status: S4 sections 4 and 18; Ryno reported no additional budget, device, connectivity or software limitations. **Engineering implication and planned response:** Evaluate future service quotas, storage, messaging and operating cost. Optional external notifications may affect both budget and reliability.

CON-010
Request data may be sensitive; security is required throughout the lifecycle.

Source/status: S4 sections 2, 4 and 16. **Engineering implication and planned response:** Define authorised visibility, least privilege, permitted status actions and sensitive-information handling. Verify denial as well as authorised use later; record residual risk.

CON-011
No prescribed stack; the team must justify capability and environment compatibility.

Source/status: S4 sections 4, 18.1 and 25. **Engineering implication and planned response:** Do not choose a stack in M1 without evidence. Assess learning curve, available devices/environment, security, testing and hosting fit before a later decision.

CON-012
Development, test, staging and production have distinct verification and operational concerns.

Source/status: S4 section 17. **Engineering implication and planned response:** Identify configuration, recovery and operational evidence needs now, for Willem's forward considerations. M1 does not implement production deployment.

4.2 Assumptions requiring validation

ASM-001
One organisation's service-request workflow is the selected initial context; no multi-tenant product commitment.

Impact if wrong: Multiple independent organisations could alter identity, isolation, reporting, costs and administration. **Owner/action:** Team confirms during formal scope review.

ASM-002
In-application feedback is the selected initial mechanism; external messaging is deferred.

Impact if wrong: A later off-platform need could require delivery retries, privacy controls and added costs. **Owner/action:** Steven specifies in-app outcomes; Willem records the deferment/risk.

ASM-003
Coordination/management is a responsibility set rather than proof of a separate organisational department.

Impact if wrong: A real distinct role could require adjusted permissions and workflow cases. **Owner/action:** Validate when an organisational representative becomes available.

ASM-004
Four hours per member per day includes meetings and reviews, and all members are available daily.

Impact if wrong: Confirmed by Ryno on 7 September 2026. **Owner/action:** Plan and record effort on this basis; update if availability changes.

ASM-005
Fictional, non-sensitive test requests will be used for early verification.

Impact if wrong: A later need for real records would introduce approval, access and data-handling constraints. **Owner/action:** Steven/Willem preserve this test-data rule and record any exception.

Except for ASM-004, these are team-selected planning positions rather than confirmed client facts. Material assumptions should be reviewed for risk treatment under S4 section 12.

4.3 Constraint interactions and trade-offs

Visibility versus sensitive information (CON-005/CON-010). SCP-003 and SCP-008 require meaningful views, while the scenario includes sensitive requests. Showing every field to every user would simplify a display but violate role-related access needs. Define requester-visible progress separately from restricted information and then derive a permission matrix. This affects requirements, data/API/UI boundaries, audit information and negative tests; the actual authentication technology remains a later decision.

Fast completion versus review capacity (CON-001/CON-002/CON-007). Each author needs the other two members' approvals. Completing Ryno's drafts early cannot by itself complete the baseline if Steven or Willem is unavailable. Reserve daily review time and resolve cross-artefact questions at the every-second-day meeting. Maintain meaningful review instead of treating approval as administrative overhead.

Feedback versus cost and delivery reliability (CON-005/CON-006/CON-009). Feedback is required. The working baseline uses in-application feedback and defers external notification services, which would add delivery failure states, usage limits, configuration and likely operating costs. If later evidence requires external delivery, assess scope, reliability, security, tests and budget together.

Overdue reporting versus workflow policy (CON-005/CON-006). The selected rule marks work overdue when its agreed target date/time has passed and it is not Resolved, Closed or Rejected; authorised staff or a manager sets the target during acceptance/assignment. Steven must make the time boundary testable and link the outcome to SCP-010/SCP-012/SCP-014 and the RTM.

Technology familiarity versus future operation (CON-003/CON-009/CON-011/CON-012). A familiar development tool may reduce initial effort but still create compatibility, hosting, recovery or service-cost problems. Keep stack selection open in M1 and gather the evidence needed for M2. The institution does not guarantee that every chosen technology will run on its environment [S4, sections 18.1 and 25].

4.4 Policy answers and remaining questions

The team can use the following working answers now and validate them with an appropriate lecturer/client representative when required.

Q-001

Use the request fields and seven categories in scope section 3.1.1; category changes are controlled by authorised management/administration.

Affected work: SCP-001/SCP-002; validation, classification and quality criteria. **Responsible follow-up:** Steven converts to requirements; team reviews.

Q-002

Requesters see their own work; authorised staff see relevant work; coordinators/managers assign and report; management has no blanket sensitive access; administration does not imply business-data access.

Affected work: SCP-006/SCP-008 to SCP-012/SCP-016/SCP-017. **Responsible follow-up:** Steven specifies permission cases; Willem reviews security risk.

Q-003

Submitted -> Accepted -> Assigned -> In Progress -> Resolved -> Closed, with authorised rejection, reassignment and requested reopening; all changes attributable.

Affected work: SCP-003/SCP-005/SCP-010/SCP-012/SCP-014. **Responsible follow-up:** Steven specifies transitions and acceptance cases.

Q-004

Overdue means the target date/time has passed and state is not Resolved, Closed or Rejected; authorised staff/manager sets target on acceptance/assignment.

Affected work: SCP-014 and reporting acceptance cases. **Responsible follow-up:** Steven makes the boundary testable.

Q-005

Provide in-application feedback for accepted, rejected, updated, resolved and closed events; external channels are deferred. Exact response-time targets remain to be agreed.

Affected work: SCP-005, FUT-004, cost/reliability/security. **Responsible follow-up:** Steven specifies behaviour; Willem records deferment.

Q-006

Apply least privilege, a sensitive-information indicator and fictional/non-sensitive test data. Exact retention period remains open because neither brief states it.

Affected work: SCP-004/SCP-008/SCP-016/SCP-017 and recovery concerns. **Responsible follow-up:** Steven/Willem preserve the open retention item.

Q-007
No extra budget/device/connectivity/software limitations are reported. Exact user volumes, operating hours and measurable performance targets remain open.

Affected work: Performance, usability, reliability and operational requirements.
Responsible follow-up: Steven proposes measurable targets for team review.

The M1 submission date is 9 September 2026. All three members have full repository access and no extra resource limitations are reported. Team-registration evidence remains unconfirmed; the Master lists 31 August 2026 as the registration deadline [S4, section 7.1]. No missed registration is inferred from the absence of evidence. No separate M1 Excel Rubric/Marking Guide has been received.

4.5 Links to requirements and risks

Steven links constraints, assumptions and policy answers to requirements and the RTM. Willem assesses risks such as review delay, ambiguous overdue policy, inappropriate data visibility and future service costs, using his required probability/impact/mitigation fields. When an assumption is validated or rejected, update this record and affected scope/requirements/risks under review; after baseline apply formal change control [S4, sections 11-14].

References: Source register, S4 sections 2-4, 6-18, 20, 23 and 25; S1 sections 3-5; S2 and S3.

5. Team Working Agreement

5.1 Purpose and responsibilities

This agreement applies the Master brief's collaboration and accountability requirements to the team's reported arrangements [S4, sections 7.1-10; S3]. The role allocation is recorded in the repository README [S2]. Ryno confirmed on 7 September 2026 that Steven and Willem agreed to these responsibilities and working arrangements.

Member	Primary M1 responsibilities	Required non-author reviewers
Ryno Goetz	Problem/business need, stakeholder analysis, scope, constraints, working agreement and GitHub governance	Steven Riaan Piek and Willem Booysen
Steven Riaan Piek	Functional/non-functional requirements, acceptance criteria and Requirements Traceability Matrix (RTM)	Ryno Goetz and Willem Booysen
Willem Booysen	Risk Register, forward considerations, Engineering Decision Log, AI Usage Register and PED integration/sign-off coordination	Ryno Goetz and Steven Riaan Piek

Each owner prepares source-supported work, responds to review and maintains its connections with other artefacts. Every member must understand the whole project, participate in presentation/defence and contribute authentic issue, commit, PR, review and documentation evidence [S4, sections 8-8.1 and 19].

5.2 Communication and capacity

- **WhatsApp:** agreed channel for short progress messages, availability and blockers.
- **Discord:** agreed channel for detailed discussions, screen sharing and scheduled meetings.
- **Daily discussions:** each member reports what changed, what comes next, dependencies and any blocked work. Summarise material decisions/actions in the relevant issue or controlled log.
- **Meetings every second day after 11:00:** inspect progress and linked artefacts, resolve policy/scope/requirement conflicts, plan reviews and record action owners. Confirm the exact start time for each meeting through the team channels.
- **Four hours per member every day:** confirmed by Ryno. Discussion, meetings, research, writing, review and corrections all count within this allowance, giving a gross planned capacity of 12 person-hours per day.

The team's WhatsApp/Discord use is for project coordination. It does not commit CivicConnect to integrating those platforms as product features.

5.3 Work and review cycle

1. Create/select an issue for each coherent work unit, with an owner, intended output, dependencies and completion criteria. Work backward from the M1 submission date of 9 September 2026 and leave time for both required reviews.
2. Work on a task branch from the latest controlled main state. Substantive direct development on main is not permitted [S4, section 9].

3. Verify sources, assumptions, identifiers, cross-links and acceptance implications before requesting review. Record material AI assistance and the human verification actually performed.
4. Open a PR linked to the issue and request both non-author members' reviews. Summarise the proposed change, rationale, affected artefacts and unresolved questions.
5. Reviewers inspect substance, including correctness, scope/requirement alignment, security/privacy, traceability and downstream verification implications. Give specific findings where correction is needed.
6. The author addresses comments, reviewers inspect corrections, and the final change receives both required independent approvals before merge. Self-approval and unexamined approval are not accepted [S4, sections 9-9.1].
7. Update related artefacts and the issue after the controlled change. Preserve the review and version evidence.

A drafted file is ready for review when it is coherent and source-supported with open assumptions visible. It is ready for baseline only when relevant questions are resolved, both reviews are complete and formal sign-off requirements are met.

5.4 Meetings, decisions and conflict resolution

Keep a short record for each actual meeting: date, attendees, artefacts examined, decisions, open questions and actions with owners/target dates. Link to issues or decision entries where practical. Do not record meetings that have not occurred.

For disagreements, compare the options against stakeholder value, required scope, evidence, capacity, quality, security and risk. Seek a reasoned agreement; record genuine significant decisions with rationale and consequences in Willem's Decision Log. If evidence is insufficient, record a justified deferment and the information needed. Ask the lecturer to resolve unclear assessment requirements; do not present internal assumptions as lecturer approval [S4, sections 12-14].

5.5 Delays and changes in availability

Report a likely missed task or review commitment promptly, identify completed work and explain the remaining dependency. Agree and record a revised handoff or redistribution. Track significant schedule variance and its engineering cause. Do not silently cut required security, testing or peer review to recover time [S4, section 18].

Each member should retain enough familiarity with the other artefacts to help resolve blockers and answer assessor questions. Reassignment does not justify invented contribution or attendance evidence.

5.6 AI, references and sensitive information

Each author remains accountable for material AI-assisted work. Record date, student, tool, engineering task, contribution, verification, decision and issues found in the team AI Usage Register. Verify important claims against credible sources or appropriate tests and be able to explain and modify the submitted work [S4, sections 10-10.1].

Use in-text citations and a reference list where claims depend on supplied or external material. Do not commit credentials or expose confidential/sensitive information to external AI systems. Actual repository configuration claims require project evidence, not a citation to this agreement [S4, sections 9-10; S1, section 9.1].

5.7 Acceptance and revision

All three members confirmed the Team Working Agreement and approved its use for Milestone 1 on 8 September 2026.

Member	Agreement status	Confirmation date
Ryno Goetz	Agreed	Confirmed and approved, 8 September 2026
Steven Riaan Piek	Agreed	Confirmed and approved, 8 September 2026
Willem Booysen	Agreed	Confirmed and approved, 8 September 2026

Review this agreement if availability or project conditions change. Controlled amendments require the same meaningful review and approval process.

References: Source register, S4 sections 7.1-10, 12-14, 18-19; S1 sections 3, 7, 9.1 and 10; S2 and S3.

6. GitHub governance and team controls

6.1 Repository and mandatory controls

The local repository's configured remote is RynoGz/SEN831_Group_F. Its tracked layout currently includes the README and Milestone_1 through Milestone_4 folders. Ryno confirmed that all three members have full repository access. This does not prove team registration or that branch controls are enabled.

GOV-001
One controlled team repository unless an alternative is justified and approved.

Governing source: S4 section 9.

GOV-002
Protect main and treat it as the controlled product state.

Governing source: S4 section 9.

GOV-003
No substantive direct development on main; use PRs for substantive changes entering it.

Governing source: S4 section 9.

GOV-004
At least two approvals from team members other than the author; self-approval is not accepted.

Governing source: S4 section 9.

GOV-005
Reviews must meaningfully inspect the work.

Governing source: S4 sections 9-9.1.

GOV-006
Meaningful engineering work appears in a project board/backlog and is linked where practical.

Governing source: S4 section 9.

GOV-007
Do not commit passwords, keys, tokens or confidential credentials.

Governing source: S4 section 9.

GOV-008
Commits, PRs, reviews, merges and changes must be progressive and authentic.

Governing source: S4 sections 9 and 23.

The main branch is protected and the team's controlled review policy requires both non-author members to approve substantive changes. The final M1 revision received both required reviews and approvals.

6.2 Author and reviewer responsibilities

Ryno Goetz
Steven Riaan Piek and Willem Booysen.

Steven Riaan Piek
Ryno Goetz and Willem Booysen.

Willem Booysen
Ryno Goetz and Steven Riaan Piek.

For Ryno's analysis, Steven should particularly check whether needs/scope produce testable, traceable requirements. Willem should particularly check risks, assumptions and decision consequences. Both reviewers inspect the complete change before approving; the focus areas do not divide their responsibility into separate partial approvals.

6.3 Document workflow

1. Represent a coherent task in the backlog, such as reviewing the requester/staff/management stakeholder analysis.
2. Create a task branch from the latest approved main state.
3. Edit Markdown source files and verify source citations, IDs, linked artefacts and declared assumptions.
4. Open a PR linked to the issue, describing the engineering purpose, affected artefacts, verification and unresolved questions.
5. Request both non-author reviewers, address findings and have the corrected content re-reviewed as necessary.
6. Merge only after the final content satisfies the required approvals and applicable controls. Link the result to the issue and update dependent work.

6.4 Task and pull request information

An issue should identify its owner, output, source/requirement links, dependencies and completion criteria. Record an agreed target date when available. A PR should link the issue and explain why the change is needed, what it affects, what was actually checked, what remains open and how material AI assistance was verified.

For documentation review, check at least these relevant questions: Does the statement match its source? Is a proposal being mistaken for an approved fact? Does scope preserve every required capability? Are identifiers consistent? Does this change affect requirements, risks or future verification? Are approvals based on the final content? For code in later milestones, extend review to implementation correctness, tests/regression, dependencies, maintainability and security [S4, section 9.1 and Appendix A].

6.5 Governance status

Main protection and review settings

Main is protected, and the team applies the Master Brief requirement of two non-author approvals.

Member access/registration

Ryno, Steven and Willem have full repository access.

Issue, branch and pull request

The M1 document change was managed through the team repository workflow.

Two independent reviews of Ryno's change

Both non-author members completed meaningful review and approval.

Controlled merge/version

PED v1.0 records the approved M1 baseline; later revisions follow the controlled change workflow.

Repository history remains the authoritative record of controlled work. M1 documentation provides valid engineering evidence before application coding or a CI pipeline is implemented [S4, section 23; S1, sections 3.1 and 5].

7. Requirements and acceptance criteria

7.1 Purpose

This document defines the baselined functional requirements (FR) and non-functional requirements (NFR) for CivicConnect for Milestone 1.

The requirements are derived from:

• the SEN381 CivicConnect Master Project Brief;

• Ryno Goetz's Member 1 stakeholder, scope, constraints and assumptions artefacts, version 0.2;

• the stakeholder needs (NEED-*) and scope capabilities (SCP-*) established by Member 1;

• requirements-engineering guidance from ISO/IEC/IEEE 29148:2018; and

• product-quality guidance from ISO/IEC 25010:2023.

The team reviewed the Member 1 foundation together with the requirements and RTM. The requirements in this PED form part of the approved v1.0 baseline.

7.2 Requirement conventions

Field	Meaning
ID	Stable requirement identifier (FR-* or NFR-*)
Source / stakeholder	Primary source and stakeholder(s) affected
Need / scope	Traceability to Member 1 NEED-* and SCP-*
Priority	Must = required minimum capability; Should = important cross-cutting quality obligation

Field	Meaning
Status	Baselined in PED v1.0; changes require control

The wording uses shall for mandatory requirements. Open policy questions are not silently answered where the Member 1 artefacts identify them as unresolved.

7.3 Functional requirements

FR-001: Submit a service request

Source / stakeholders: S4 section 3; STK-001 Requesters; STK-002 Authorised service staff
Need / scope: NEED-001 / SCP-001
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall permit a requester to submit a new service request with a title, description, category, location, requester identity/contact information, and a sensitive-information indicator. The system shall automatically record the submission date/time and allow an optional attachment.

Acceptance Criteria:

1. A submission is rejected when any required field is missing.

2. A successfully submitted request contains all required request information.

3. The submission date/time is recorded automatically rather than entered manually by the requester.

4. Omitting an attachment does not prevent a valid request from being submitted.

5. After successful submission, the request is recorded in the Submitted state.

6. The M1 capability does not require live ingestion from WhatsApp, email, telephone, or paper workflows.

FR-002: Controlled request categorisation

Source / stakeholders: S4 section 3; STK-001 Requesters; STK-004 Service coordination
Need / scope: NEED-002 / SCP-002
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall provide a controlled category mechanism using the initial categories defined in the Member 1 working scope baseline: Maintenance, IT support, Facility fault, Damaged equipment, Security concern, Lost property, and Other.

Acceptance Criteria:

1. A requester selects a category from the available controlled category set.

2. A submitted request stores the selected category.

3. The initial category set contains the seven categories defined in the Member 1 working scope baseline.

4. A request cannot use an uncontrolled free-text category value instead of a controlled category.

5. The M1 requirement does not require AI-based automatic classification.

FR-003: View current request status

Source / stakeholders: S4 section 3; STK-001 Requesters
Need / scope: NEED-003 / SCP-003
Priority: Must
Status: Baselined in PED v1.0

Requirement:
An authorised requester shall be able to view the status of their submitted service requests through CivicConnect.

Acceptance Criteria:

1. An authorised requester can view the status of a request they are permitted to access.
2. The displayed status matches the recorded lifecycle state of the request.
3. A requester cannot access another request through the status view without authorisation.
4. Status values shown to the requester use the controlled lifecycle terminology.

FR-004: View request history/list

Source / stakeholders: S4 section 3; STK-001 Requesters
Need / scope: NEED-003 / SCP-004
Priority: Must
Status: Baselined in PED v1.0

Requirement:
Upon confirmation of the retention policy, CivicConnect shall permit an authorised requester to view a retrievable list/history of their previously submitted service requests.

Acceptance Criteria:

1. An authorised requester can retrieve requests they are permitted to view.
2. The history identifies each request sufficiently to distinguish it from other requests.
3. Historic requests preserve their recorded lifecycle information.
4. The system does not claim infinite storage; the final retention period remains subject to confirmation.

FR-005: Provide in-application request feedback

Source / stakeholders: S4 section 3; STK-001 Requesters
Need / scope: NEED-003 / SCP-005
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall provide the requester with meaningful in-application feedback when a request is accepted, rejected, updated, resolved, or closed.

Acceptance Criteria:

1. The requester can identify when their request has been accepted.
2. The requester can identify when their request has been rejected.
3. The requester can identify relevant updates to their request.
4. The requester can identify when their request is resolved.

5. The requester can identify when their request is closed.

6. The M1 requirement does not require SMS, WhatsApp, or email notification delivery.

FR-006: View work relevant to authorised staff

Source / stakeholders: S4 section 3; STK-002 Authorised service staff
Need / scope: NEED-004, NEED-009 / SCP-006
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall allow authorised service staff to access service requests consistent with their duties and permissions.

Acceptance Criteria:

1. An authorised staff user can view requests within the scope of their permitted work.

2. A staff member cannot automatically view all requests merely because they are staff.

3. Access decisions are based on the agreed responsibility and permission model.

4. Sensitive requests remain subject to the applicable access rules.

FR-007: Search, filter and sort requests

Source / stakeholders: S4 section 3; STK-002 Authorised service staff
Need / scope: NEED-004 / SCP-007
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall permit authorised staff to search, filter, and/or sort service requests according to useful criteria agreed for the operational workflow.

Acceptance Criteria:

1. An authorised staff member can retrieve relevant requests using the agreed search, filter, and/or sort capabilities.

2. Search and filtering apply only to requests the user is authorised to access.

3. The returned results match the selected search, filter, or sort criteria.

4. Exact operational criteria are confirmed during requirements review rather than invented as an M1 fact.

5. A sophisticated search engine is not required unless the team justifies it.

FR-008: View authorised request details

Source / stakeholders: S4 section 3; STK-002 Authorised service staff
Need / scope: NEED-004, NEED-009 / SCP-008
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall allow an authorised user to view request details within the scope of their responsibility and access rights, including sensitive information only where authorised.

Acceptance Criteria:

1. An authorised user can view request information permitted by their role or responsibility.

2. Sensitive information is not displayed to a user without the required authorisation.

3. The detailed view includes the request information captured by FR-001, subject to the user’s permissions.

4. Administrative rights do not automatically grant access to business request details.

FR-009: Assign, accept and reassign responsibility

Source / stakeholders: S4 sections 2–3; STK-002 Authorised service staff; STK-004 Service coordination
Need / scope: NEED-005 / SCP-009
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall allow authorised staff to accept or be assigned responsibility for a service request, and shall allow authorised coordinators or managers to assign or reassign responsibility.

Acceptance Criteria:

1. An authorised staff user can accept responsibility where permitted.

2. An authorised coordinator or manager can assign responsibility to an appropriate staff user.

3. An authorised coordinator or manager can reassign responsibility where permitted.

4. A user without the required permission cannot assign or reassign responsibility.

5. The current responsible user or responsibility state is identifiable for the request.

6. When responsibility changes from an existing staff member to a different staff member, a valid reason is required and is retained with the assignee, acting user and timestamp. First assignment remains optional. Approved under CR-M3-01 on 15 October 2026.

FR-010: Enforce controlled request status transitions

Source / stakeholders: S4 section 3; STK-002 Authorised service staff; STK-004 Service coordination
Need / scope: NEED-006 / SCP-010
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall enforce the working request lifecycle Submitted → Accepted → Assigned → In Progress → Resolved → Closed, with controlled rejection from Submitted, authorised reassignment, and a controlled mechanism for requesting reopening.

Acceptance Criteria:

1. A newly submitted request starts in the Submitted state.

2. A request can transition only through transitions permitted by the agreed lifecycle rules.

3. An authorised user can reject a request only when it is in the Submitted state and the user’s permissions allow rejection.

4. Assignment and reassignment are permitted only for authorised users.

5. A status change records the user responsible for the change.

6. An unauthorised or invalid status transition is blocked.

7. Reopening follows the approved working policy recorded under Q-003; later refinements require controlled change.

FR-011: Record actions, comments and resolution information

Source / stakeholders: S4 section 3; STK-002 Authorised service staff
Need / scope: NEED-006 / SCP-011
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall permit authorised staff to record relevant actions, comments, and resolution information against a service request, with recorded information controlled for its intended visibility.

Acceptance Criteria:

1. An authorised user can record a relevant action or comment against a request.

2. An authorised user can record resolution information when resolving work.

3. Recorded entries remain associated with the correct request.

4. Where the final policy requires it, the system distinguishes requester-visible information from information restricted to authorised users.

5. Accountable actions have a responsible user attached to them.

FR-012: Resolve or close requests when authorised

Source / stakeholders: S4 section 3; STK-002 Authorised service staff; STK-004 Service coordination
Need / scope: NEED-006 / SCP-012
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall permit only authorised users to resolve or close service requests in accordance with the controlled request lifecycle and agreed responsibilities.

Acceptance Criteria:

1. An authorised user can move a request to Resolved when the agreed resolution conditions are met.

2. An authorised user can move a request to Closed when the agreed closure conditions are met.

3. An unauthorised user cannot resolve or close a request.

4. The system records the user responsible for the resolution or closure action.

5. The approved working distinction between Resolved and Closed applies; later refinements to reopening rules require controlled change.

FR-013: View service activity

Source / stakeholders: S4 section 3; STK-003 Management/oversight
Need / scope: NEED-007 / SCP-013
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall provide authorised management/oversight users with sufficient service activity information to monitor the request workflow.

Acceptance Criteria:

1. An authorised management/oversight user can view organisational service activity within their permitted scope.

2. The activity information includes recorded requests and relevant lifecycle information.

3. Access to the activity view is restricted according to the agreed permission model.

4. The management view is validated against representative request scenarios during later design and testing.

FR-014: Identify open, overdue, resolved and closed work

Source / stakeholders: S4 section 3; STK-003 Management/oversight
Need / scope: NEED-007 / SCP-014
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall permit authorised users to identify open, overdue, resolved, and closed service requests using the agreed lifecycle and overdue definitions.

Acceptance Criteria:

1. Requests in the relevant open states can be identified separately from resolved, closed, and rejected requests.

2. Open work can be identified from the controlled lifecycle state.

3. A request is considered overdue when its agreed target date/time has passed and its status is not Resolved, Closed, or Rejected.

4. The target date/time is set by an authorised staff member or manager according to the working policy.

5. Overdue classification is verified using a controlled test case at or around the target date/time boundary.

FR-015: Analyse requests by category, status and justified dimensions

Source / stakeholders: S4 section 3; STK-003 Management/oversight
Need / scope: NEED-008 / SCP-015
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall permit authorised users to analyse requests by category and status, and, where justified, by other agreed dimensions.

Acceptance Criteria:

1. Authorised management users can group, filter, or otherwise present request information by category.

2. Authorised management users can analyse request information by lifecycle status.

3. Any additional analysis dimension is included only when its operational value is justified and agreed.

4. The analysis respects the visibility permissions of the authorised user.

5. Predictive analytics are not required for M1.

FR-016: Support accountability and service-performance analysis
Source / stakeholders: S4 section 3; STK-003 Management/oversight
Need / scope: NEED-008 / SCP-016
Priority: Must
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall retain sufficient information about attributable requests, responsibilities, and progress to support accountability and service-performance analysis.

Acceptance Criteria:

1. The system retains sufficient lifecycle history to determine how a request progressed.

2. Responsibility for relevant work can be identified from recorded request information.

3. Status changes can be attributed to the user who made them.

4. Management can explain request progress and responsibility using recorded information.

5. Predictive analytics and a data warehouse are not required for M1.

7.4 Non-functional requirements

NFR-001: Authorisation and least-privilege access
Source / stakeholders: S4 sections 2, 4 and 16; STK-001, STK-002, STK-003, STK-005
Need / scope: NEED-009 / SCP-017
Priority: Should
Status: Baselined in PED v1.0

Requirement:
Using the least-privilege approach, CivicConnect shall limit request access and permitted actions according to authorised responsibilities.

Acceptance Criteria:

1. A requester can view their own requests in accordance with the agreed visibility policy.

2. Staff can view work relating to their authorised role or responsibility.

3. Only authorised users can assign, reassign, change status, resolve, or close requests.

4. Access to management reporting is permission-controlled.

5. Administrative privileges do not automatically grant business-data access.

NFR-002: Sensitive-information protection
Source / stakeholders: S4 sections 2, 4 and 16; STK-001, STK-002, STK-005
Need / scope: NEED-009 / SCP-017
Priority: Should
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall prevent unauthorised disclosure of sensitive request information throughout the request lifecycle.

Acceptance Criteria:

1. Sensitive information is identified using the agreed sensitive-information indicator and protected according to the applicable access rules.

2. Only authorised users can view sensitive request information.

3. Unauthorised attempts to access protected information are denied.

4. Verification uses fictional, non-sensitive test data.

5. The final retention period is not invented in M1 and remains subject to confirmation.

NFR-003: Auditability and accountability

Source / stakeholders: S4 sections 3, 8 and 11; STK-002, STK-003, STK-006
Need / scope: NEED-006, NEED-008 / SCP-010, SCP-011, SCP-016
Priority: Should
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall maintain attributable records of relevant lifecycle changes, responsibility changes, and actions so that request history can be reviewed for accountability purposes.

Acceptance Criteria:

1. Each status change identifies the resulting status and responsible user.

2. Assignments and reassignments are attributable to the user who performed them.

3. Relevant actions, comments, and resolution information remain associated with the request.

4. The resulting history is sufficient to reconstruct how the request progressed during verification.

NFR-004: Usability and understandable feedback

Source / stakeholders: S4 sections 3 and 15; STK-001, STK-002, STK-003
Need / scope: NEED-001, NEED-003, NEED-004 / SCP-001, SCP-003 to SCP-007, SCP-013 to SCP-016
Priority: Should
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall present request information, status, and feedback in a form that the intended user can understand and use for the supported task.

Acceptance Criteria:

1. Requesters can identify the current status of their own requests during representative task-based verification.

2. Feedback identifies the relevant request event or outcome.

3. Staff can identify relevant work by following an agreed search, filter, and/or sort workflow.

4. Management can interpret the status and activity information used for oversight.

5. Usability is supported by defined verification tasks and criteria rather than an unsupported claim that the system is “user-friendly.”

NFR-005: Data integrity and consistency

Source / stakeholders: S4 sections 3, 11 and 15; STK-002, STK-003
Need / scope: NEED-001, NEED-005, NEED-006, NEED-008 / SCP-001, SCP-009 to SCP-

016
Priority: Should
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall maintain consistency between recorded request details, responsibility, lifecycle state, and attributable history.

Acceptance Criteria:

1. Each request has one current lifecycle state at a time.

2. Changes in responsibility are recorded consistently.

3. The status history reflects the status changes that occurred.

4. An unauthorised or invalid transition does not leave the request in an inconsistent lifecycle state.

5. Repeated and boundary cases are included in later verification.

NFR-006: Performance under agreed operating conditions

Source / stakeholders: S4 sections 15 and 18; STK-003, STK-005, STK-006
Need / scope: NEED-010 / SCP-018
Priority: Should
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall support responsive request submission, retrieval, search, filtering, sorting, and oversight operations under the agreed operating conditions and workload.

Acceptance Criteria:

1. Working M1 performance target: using 1,000 seeded service requests and three concurrent representative users, at least 95% of request retrieval, search, filter, sort and management-view operations complete within two seconds.

2. Under the same test conditions, at least 95% of valid request submissions complete and return acknowledgement within three seconds.

3. Verification evidence records the environment, dataset, concurrency, operation, sample count, measured response times and pass/fail result.

4. The team reviews these working targets during architecture and deployment planning; any approved change is recorded through change control.

NFR-007: Reliability and recoverability

Source / stakeholders: S4 sections 15, 17 and 18; STK-005, STK-006
Need / scope: NEED-010 / SCP-018
Priority: Should
Status: Baselined in PED v1.0

Requirement:
CivicConnect shall support controlled recovery of service-request information and configuration appropriate to the agreed operating environment.

Acceptance Criteria:

1. Service-request records, status history, assignments and required configuration are included in the defined backup scope.

2. A recovery test restores the agreed backup into a controlled environment within 30 minutes.

3. The recovery test demonstrates a maximum planned data-loss window of 24 hours and records the backup time, restore time, environment and result.

4. The 30-minute recovery and 24-hour data-loss targets are working M1 thresholds and must be reviewed against the selected platform before the v2.0 architecture baseline.

NFR-008: Maintainability and controlled change
Source / stakeholders: S4 sections 6, 8, 11 and 18; STK-006 Project team
Need / scope: NEED-010, NEED-011 / SCP-018
Priority: Should
Status: Baselined in PED v1.0

Requirement:
Using requirement identifiers, traceability, version history, and verification evidence, CivicConnect and its engineering artefacts shall support controlled change.

Acceptance Criteria:

1. Requirements retain stable identifiers unless a controlled change explicitly requires a new identifier.

2. Approved requirement changes are reflected in the RTM and affected artefacts.

3. Design, implementation, and test evidence can later be associated with the applicable requirement.

4. Material changes follow the team-controlled change process.

5. Repository history records substantial changes without silently overwriting approved evidence.

7.5 Open questions and controlled deferrals

The following client-policy items remain controlled deferrals because the supplied briefs do not determine them. Any later resolution must follow change control.

Open question	Affected requirements	Current control
Q-001: Exact required request information, categories, and who may change them	FR-001, FR-002, FR-007	Working policy approved for v1.0; refine through controlled change if stakeholder evidence differs
Q-002: Final roles, visibility, assignment/reassignment, resolve/close authority	FR-006, FR-008, FR-009, FR-012, NFR-001	Least-privilege policy approved for v1.0; refine the permission matrix through controlled change
Q-003: Detailed state-transition, rejection, resolved/closed, and reopening policy	FR-010, FR-012	Lifecycle policy approved for v1.0; refine detailed transitions through controlled change
Q-004: Final overdue definition, time basis, and state treatment	FR-014	Working overdue definition approved; verify boundary cases during later design and testing
Q-005: Final feedback timing, content, and mechanism	FR-005	In-application feedback approved; refine content and timing through controlled change
Q-006: Sensitive-information handling and retention policy	FR-004, FR-008, NFR-002	Least privilege and non-sensitive test data approved; retention remains a controlled deferral

Open question	Affected requirements	Current control
Q-007: User volumes, operating hours, devices/connectivity, and other conditions needed to define measurable quality criteria	NFR-004, NFR-006, NFR-007	M1 performance and recovery targets approved; refine operating conditions through controlled change

The M1 performance and recovery targets baseline NFR-006 and NFR-007. Client-specific workload, retention and operating constraints may refine them later only through controlled review.

7.6 Requirements quality basis

The requirements are intended to be clear, uniquely identifiable, traceable to sources/stakeholders, and verifiable. ISO/IEC/IEEE 29148:2018 provides the requirements-engineering basis for requirements processes and information items. ISO/IEC 25010:2023 provides the product-quality model used to structure the cross-cutting quality requirements.

The project Master Brief remains the authoritative source for project-specific requirements and controls. The ISO standards are supporting engineering references and do not override the assignment brief.

8. Requirements traceability matrix M1 baseline snapshot

8.1 Purpose

This Requirements Traceability Matrix (RTM) provides the controlled link between the Member 1 stakeholder/source analysis, stakeholder needs, scope capabilities, requirements, and acceptance criteria.

Later lifecycle evidence is intentionally marked Not evidenced at M1 because M1 does not yet contain the final architecture, implementation, tests, or release evidence.

The intended lifecycle trace is:

Stakeholder/Source → Need → Scope Capability → Requirement → Acceptance Criteria → Design → Issue/PR → Implementation → Test → Acceptance/Release Evidence

8.2 Traceability matrix

Source / Stakeholder	Need	Scope	Requirement	Acceptance criteria	Design	Issue / PR	Implementation	Test	Acceptance / Release
STK-001 / S4	NEED-001	SCP-001	FR-001	FR-001 AC1–AC6	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-001, STK-004 / S4	NEED-002	SCP-002	FR-002	FR-002 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-001 / S4	NEED-003	SCP-003	FR-003	FR-003 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-001 / S4	NEED-003	SCP-004	FR-004	FR-004 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-001 / S4	NEED-003	SCP-005	FR-005	FR-005 AC1–AC6	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002 / S4	NEED-004, NEED-009	SCP-006	FR-006	FR-006 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002 / S4	NEED-004	SCP-007	FR-007	FR-007 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002 / S4	NEED-004, NEED-009	SCP-008	FR-008	FR-008 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002, STK-004 / S4	NEED-005	SCP-009	FR-009	FR-009 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002, STK-004 / S4	NEED-006	SCP-010	FR-010	FR-010 AC1–AC7	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002 / S4	NEED-006	SCP-011	FR-011	FR-011 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002, STK-004 / S4	NEED-006	SCP-012	FR-012	FR-012 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-003 / S4	NEED-007	SCP-013	FR-013	FR-013 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-003 / S4	NEED-007	SCP-014	FR-014	FR-014 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-003 / S4	NEED-008	SCP-015	FR-015	FR-015 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1

Source / Stakeholder	Need	Scope	Requirement	Acceptance criteria	Design	Issue / PR	Implementation	Test	Acceptance / Release
STK-003 / S4	NEED-008	SCP-016	FR-016	FR-016 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-001, STK-002, STK-003, STK-005 / S4	NEED-009	SCP-017	NFR-001	NFR-001 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-001, STK-002, STK-005 / S4	NEED-009	SCP-017	NFR-002	NFR-002 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002, STK-003, STK-006 / S4	NEED-006, NEED-008	SCP-010, SCP-011, SCP-016	NFR-003	NFR-003 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-001, STK-002, STK-003 / S4	NEED-001, NEED-003, NEED-004	SCP-001, SCP-003–SCP-007, SCP-013–SCP-016	NFR-004	NFR-004 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-002, STK-003 / S4	NEED-001, NEED-005, NEED-006, NEED-008	SCP-001, SCP-009–SCP-016	NFR-005	NFR-005 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-003, STK-005, STK-006 / S4	NEED-010	SCP-018	NFR-006	NFR-006 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-005, STK-006 / S4	NEED-010	SCP-018	NFR-007	NFR-007 AC1–AC4	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1
STK-006 / S4	NEED-010, NEED-011	SCP-018	NFR-008	NFR-008 AC1–AC5	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1	Not evidenced at M1

8.3 Requirement-to-source coverage

Requirement group	Covered source capabilities
FR-001	SCP-001 / NEED-001
FR-002	SCP-002 / NEED-002
FR-003	SCP-003 / NEED-003
FR-004	SCP-004 / NEED-003
FR-005	SCP-005 / NEED-003
FR-006	SCP-006 / NEED-004, NEED-009
FR-007	SCP-007 / NEED-004
FR-008	SCP-008 / NEED-004, NEED-009
FR-009	SCP-009 / NEED-005
FR-010	SCP-010 / NEED-006
FR-011	SCP-011 / NEED-006
FR-012	SCP-012 / NEED-006
FR-013	SCP-013 / NEED-007
FR-014	SCP-014 / NEED-007
FR-015	SCP-015 / NEED-008
FR-016	SCP-016 / NEED-008
NFR-001, NFR-002	SCP-017 / NEED-009
NFR-003	SCP-010, SCP-011, SCP-016 / NEED-006, NEED-008
NFR-004	SCP-001, SCP-003–SCP-007, SCP-013–SCP-016
NFR-005	SCP-001, SCP-009–SCP-016
NFR-006, NFR-007, NFR-008	SCP-018 / NEED-010, NEED-011

8.4 Open-question traceability

Open question	Affected requirements	Current control
Q-001: Exact request information, categories, and category-change authority	FR-001, FR-002, FR-007	Working policy approved for v1.0; refine through controlled change if stakeholder evidence differs
Q-002: Final roles, visibility, assignment/reassignment, resolve/close authority	FR-006, FR-008, FR-009, FR-012, NFR-001	Least-privilege policy approved for v1.0; refine the permission matrix through controlled change
Q-003: Detailed transitions, rejection, resolved/closed, and reopening	FR-010, FR-012	Lifecycle policy approved for v1.0; refine detailed transitions through controlled change
Q-004: Overdue time basis and state treatment	FR-014	Working overdue definition approved; verify boundary cases during later design and testing

Open question	Affected requirements	Current control
Q-005: Feedback timing, content, and mechanism	FR-005	In-application feedback approved; refine content and timing through controlled change
Q-006: Sensitive-information handling and retention	FR-004, FR-008, NFR-002	Least privilege and non-sensitive test data approved; retention remains a controlled deferral
Q-007: User volumes, operating hours, devices/connectivity, and other conditions needed for measurable quality criteria	NFR-004, NFR-006, NFR-007	M1 performance and recovery targets approved; refine operating conditions through controlled change

8.5 Later-lifecycle evidence

The following columns remain intentionally open in M1:

• Design: populated when architecture/design decisions are made.

• Issue / PR: link the issue and PR that implement or change the requirement.

• Implementation: link the relevant code, configuration, or documentation evidence.

• Test: link test cases/results demonstrating the acceptance criteria.

• Acceptance / Release: link approved baseline/release evidence.

No later evidence is claimed before it exists.

8.6 Traceability control rules

• Requirement IDs remain stable unless a controlled change explicitly requires otherwise.

• Any approved requirement change updates this RTM and affected artefacts.

• Acceptance criteria remain linked to the requirement they verify.

• A requirement must not be marked complete merely because it has been designed or implemented; verification evidence is required.

• Later design, implementation, and test evidence must link back to the applicable requirement ID.

• Open questions remain visible until resolved through the team’s change/review process.

• The RTM forms part of the evolving PED and must preserve its version history.

8.7 Status summary

Category	Count	Status
Functional requirements	16	Baselined
Non-functional requirements	8	Baselined
Total requirements	24	Baselined
Scope capabilities SCP-001–SCP-016	16	Covered
Cross-cutting obligations SCP-017–SCP-018	2	Covered
Later design/implementation/test/release evidence	,	Not evidenced at M1

9. Initial risk register M1 baseline snapshot

The register records current M1 risks. Owners coordinate treatment but do not remove collective responsibility. Status remains Open until the mitigation is evidenced or the risk is formally closed.

ID	Description	Cause	Consequence	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
R1	Scope creep from new stakeholder demands	Features requested after baseline	Delay, missed deadlines and increased cost	Medium	High	High	Defend baseline and perform impact analysis	Reprioritise or formally adjust scope and schedule	Willem Booysen	Open
R2	GitHub governance not enforced	Protection or review controls incomplete	Uncontrolled changes and weak traceability	Medium	High	High	Require protected main and two non-author approvals	Audit settings and stop non-compliant merges	Ryno Goetz	Open
R3	Security requirements overlooked	Sensitive-data and access rules remain incomplete	Disclosure, redesign or compliance failure	Medium	High	High	Baseline least-privilege and sensitive-data requirements	Security review and corrective change request	Steven Riaan Piek	Open
R4	Forward-engineering concerns ignored	Later concerns are deferred without criteria	Technical debt and constrained design options	Medium	Medium	Medium	Maintain forward-considerations and decision evidence	Revisit before M2 decisions	Willem Booysen	Open
R5	Weak PED document control	Version history or controlled changes become incomplete	Assessment penalty and an unreliable baseline	Low	High	Medium	Maintain version history and require team review for controlled changes	Issue a corrected version and repeat the affected review	All members	Open
R6	Schedule slippage	Effort underestimated or feedback delayed	Rushed work and missed milestone evidence	Medium	High	High	Daily tracking, review capacity and buffer time	Reallocate work and control scope	Ryno Goetz	Open
R7	Team governance breakdown	Poor accountability or collaboration	Missed artefacts and weak individual evidence	Medium	Medium	Medium	Working agreement, issues and regular check-ins	Escalate unresolved breakdown to lecturer	All members	Open
R8	Deployment environment mismatch	Client infrastructure constraints are unknown	Rework or failed deployment	Medium	High	High	Confirm environment constraints before stack selection	Use a compatible alternative and update ADR	Willem Booysen	Open

ID	Description	Cause	Consequence	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
R9	Dependency or service-limit failure	External libraries or free tiers are not evaluated	Build failure, vulnerability or unexpected cost	Medium	High	High	Record dependencies, maturity, limits and likely cost	Replace, patch or select an approved alternative	All members	Open
R10	AI accountability failure	Material AI use is incomplete or insufficiently verified	Integrity penalty or indefensible artefact	Low	High	High	Maintain required AI fields and retain human verification	Correct or withdraw unsupported material	All members	Open

9.1 Highest concern risk

R1 remains the highest immediate concern because it can trigger schedule, quality, cost and review failures simultaneously. R2, R3, R6, R8, R9 and R10 also carry High priority. R1 is selected as the lead concern by tie-break because stakeholder pressure can arise immediately after baseline and can create cascading effects across several other risks. The response is to defend the agreed scope and require formal impact analysis before accepting additions [S1, section 3; S4, sections 4, 12 and 14].

10. Forward engineering considerations M1 baseline snapshot

Concern	Why it matters now	Later activity affected	Missing information	Risk of ignoring
Security	Protects sensitive request information and access boundaries	Architecture, identity and deployment	Applicable privacy rules and final permission matrix	Costly redesign or disclosure
Testability	Requirements and acceptance criteria must support verification	Test strategy, automation and CI	Architecture, stack and risk-based test priorities	Late defects and weak evidence
Deployment environment	Environment limits can constrain technology	Hosting and production-like staging	Client infrastructure, connectivity and support constraints	Rework or deployment failure
Data migration and recovery	Existing fragmented records may need controlled treatment	Persistence, migration, backup and restore	Source formats, quality, retention and ownership	Data loss, delay or failed migration
Observability	Failures and overdue work must be detectable	Logging, monitoring and incident response	Operational metrics, alert ownership and tooling	Failures remain hidden
Maintainability	A small team must support controlled future change	Modularity, standards and technical-debt control	Final architecture and team capability evidence	Costly change and accumulated debt
Operational cost	The solution must remain sustainable beyond the classroom	Platform, scaling and support decisions	Service pricing, free-tier limits and expected usage	Unsustainable or locked-in solution

These seven concerns preserve later options without selecting a final stack, architecture, database, testing framework or deployment platform in M1 [S1, sections 4-5; S4, sections 15-18].

11. Engineering decision log M1 baseline snapshot

ID	Context and constraints	Alternatives	Decision and rationale	Trade-offs and risks	Evidence	Deferred?	Later consequence
D1	Three-person team, M1 schedule and mandatory web workflow	Include mobile application scope now	Exclude native mobile applications from the initial baseline; web delivery covers the minimum capabilities with lower workload	Less mobile-specific access; avoids duplicated design, testing and deployment	Scope baseline and stakeholder review	No	Reassess only through change control
D2	The Master Brief requires protected main, Pull Requests and two non-author approvals	One approval; no branch protection	Apply protected main and require both non-author members to approve substantive changes	Review takes time but strengthens accountability and baseline integrity	Protected-main configuration and completed M1 reviews	No	Every substantive author depends on both other reviewers
D3	The integrated PED, reviews and M1 registers are complete	Retain separate member reports; delay approval	Approve the unified PED v1.0 as the controlled M1 engineering baseline	Creates a stable foundation; later changes require formal control	Version history, team review and approved PED	No	M2 work must use this baseline and record approved changes
D4	M1 is not a technology-selection milestone and client constraints are incomplete	Choose a familiar stack now	Defer final stack selection until M2 evidence is available	Preserves flexibility; delays implementation-specific planning	Requirements stability, capability, environment, security and cost research	Yes	M2 must compare viable alternatives
D5	Acceptance criteria exist, but stack, architecture and automation environment are not selected	Select a test framework now	Defer the framework choice, not testability or acceptance criteria	Avoids premature tool lock-in; requires M2 follow-up	Architecture, stack and risk-based test strategy	Yes	M2 maps test techniques and tools to requirements and risks
D6	Stakeholder needs may conflict and no single stakeholder has unlimited authority	Ignore conflict; escalate every disagreement	Use evidence-based negotiation and record significant unresolved decisions	Discussion consumes time; improves transparency and trust	Meeting notes, issue discussion and affected artefacts	No	Conflicts remain traceable to scope and requirements
D7	AI is permitted only with disclosure, verification and student accountability	Ban AI; accept AI output without verification	Allow material AI assistance when it is recorded and independently checked	Faster drafting but added verification and integrity risk	AI Usage Register and reviewed artefacts	No	Students must explain and modify all submitted work

12. AI usage register M1 baseline snapshot

Material AI assistance is recorded below together with the human verification, decisions and issues identified by each responsible student.

Date	Student	Tool	Engineering task	AI contribution	Human verification	Decision	Issues found
7-8 Sep 2026	Ryno Goetz	OpenAI Codex	Foundation and governance documentation	Initial drafting, restructuring and consistency review	Checked against S1/S4 and corrected project-specific assumptions and scope	Accepted with corrections	Removed unsupported completion claims and clarified controlled deferrals
8 Sep 2026	Ryno Goetz	OpenAI Codex	Table visualisation	Converted register data into readable standalone table layouts	Compared every row and field with the source register	Accepted after verification	Corrected hidden merged-cell text and layout
8 Sep 2026	Ryno Goetz	OpenAI Codex	Unified PED quality review	Compared all member parts and proposed corrections	Rechecked findings against S1/S4 and verified the final corrections	Accepted with corrections	Found register fields, count and cross-document inconsistencies
8 Sep 2026	Steven Riaan Piek	ChatGPT	Requirements wording	Suggested clearer requirement and acceptance-criteria wording	Compared wording with the minimum capabilities, Member 1 scope and S1/S4	Accepted with corrections	Improved clarity and testability; wording was manually corrected
8 Sep 2026	Steven Riaan Piek	ChatGPT	RTM organisation	Suggested matrix structure and traceability columns	Checked all FR/NFR, stakeholder, need, scope and acceptance links	Accepted with corrections	Identified missing later-lifecycle links, retained as Not evidenced at M1 for future milestones
8 Sep 2026	Willem Booysen	Copilot	Risk prioritisation	Suggested R1 as the highest immediate concern and drafted reasoning	Compared R1 with other High risks using probability, impact, immediacy and cascading effects	Accepted with corrections	R1 tied by priority; tie-break rationale added

13 Milestone 2 baseline overview M2 baseline snapshot

Milestone 2 extends PED v1.0 with architecture, persistence, technology and detailed design decisions and with traceable application evidence. The governing requirement baseline remains the 16 functional and eight non-functional requirements approved in M1. No requirement wording or identifier changed during the M2 review. Sections 13 to 18 are the 30 September snapshot. Their deferred capabilities and verification results describe M2; the current M3 implementation and results are recorded in sections 24 to 32.

13.1 M1 requirements review outcome

Review area	M2 outcome	Control
FR and NFR wording	All 24 requirements retained without change	M1 wording and acceptance criteria remain authoritative
First functional slice	NEED-001 to SCP-001 to FR-001 selected for initial development	Selection does not remove remaining scope
Controlled categories	Seven M1 categories retained	Database constraint and seed data use the approved names
Lifecycle	Seven approved statuses retained; detailed forward transitions selected	Reopening remains deferred pending controlled refinement
Open policy items	Retention, detailed role matrix, attachments and deployment evidence remain visible	No missing decision is presented as implemented

13.2 Team contributions and approvals

Member	Primary M2 contribution	Integrated evidence	Approval status
Ryno Goetz	Architecture, technology alignment, Next.js setup, frontend and README	PR 22; commit 9d84b9c; Member 1 v0.2 and frontend handoff	Approved
Steven Riaan Piek	Requirements review, data model, persistence, migrations, authentication and request path	PR 24; commits 4aad214 and d10ad28; RTM v0.4	Approved
Willem Booysen	Detailed design decisions, registers and request transition policy	PRs 23 and 25; commits 4e08af2 and 622d4c9	Approved

13.3 M2 baseline conclusion

The repository contains a project-specific Next.js application, versioned Supabase schema and access policies, requester submission and retrieval code, and automated frontend-validation and transition-policy tests. The team has not completed the full CivicConnect product. Attachments, staff workflows, later status operations, management reporting, quantitative performance evidence and restore evidence remain later work under the unchanged M1 requirements.

14 Architecture and quality drivers M2 baseline snapshot

14.1 Architecture significant requirements

ASR	Driver	Trace	Architectural response
RG-ASR-01	Authorised access	FR-003, FR-006, FR-008, FR-009, FR-012; NFR-001; R3	Derive identity from the session and enforce protected operations on the server and in database policies.
RG-ASR-02	Sensitive information	FR-008; NFR-002; R3	Return only permitted fields, keep attachment storage private and keep privileged secrets outside browser code.
RG-ASR-03	Consistency and attribution	FR-010 to FR-012, FR-016; NFR-003, NFR-005	Centralise lifecycle rules and keep request and history writes atomic and attributable.
RG-ASR-04	Responsive operation	NFR-006; R8, R9	Use one application boundary, index relevant queries and measure against the approved workload.
RG-ASR-05	Recoverability	NFR-007; R8, R9	Treat database, attachment objects and configuration as separate recovery concerns and perform a timed restore later.
RG-ASR-06	Maintainability	NFR-008; CON-001, CON-002, CON-007; R6	Use clear internal responsibilities, one JavaScript codebase, a lockfile and controlled reviews.
RG-ASR-07	Understandable interaction	FR-001, FR-003 to FR-005; NFR-004	Use labelled controls, field feedback and confirmation only after a save succeeds.

NFR-006 retains the approved workload of 1,000 seeded requests and three concurrent representative users, with 95 percent of reads completing within two seconds and 95 percent of valid submissions returning acknowledgement within three seconds. NFR-007 retains recovery within 30 minutes and a maximum planned data-loss window of 24 hours. M2 records these as targets, not achieved results.

14.2 Architecture decision

Alternative	Benefit	Trade off	Decision
Simple layered application	Small setup and familiar presentation, business and data separation	Horizontal layers can mix request, access and infrastructure rules	Viable but not selected
Modular monolith with internal layers	One codebase and release boundary with explicit responsibilities	Requires discipline to maintain boundaries	Selected
Independently deployed services	Independent deployment and scaling	Adds network contracts, distributed failure and consistency work	Not proportionate for the team or current load

RG-ADR-01 is accepted. CivicConnect is organised as a modular monolith within one Next.js application. Presentation, server entry points, request operations, lifecycle rules and infrastructure adapters remain distinct logical responsibilities while sharing one deployment boundary. Durable state belongs in Supabase PostgreSQL or private Storage, not process memory or a deployment filesystem.

Local development and any hosted preview must use fictional data and controlled non-production configuration. Vercel preview deployment and production release remain separate readiness claims. No production deployment is claimed in this baseline.

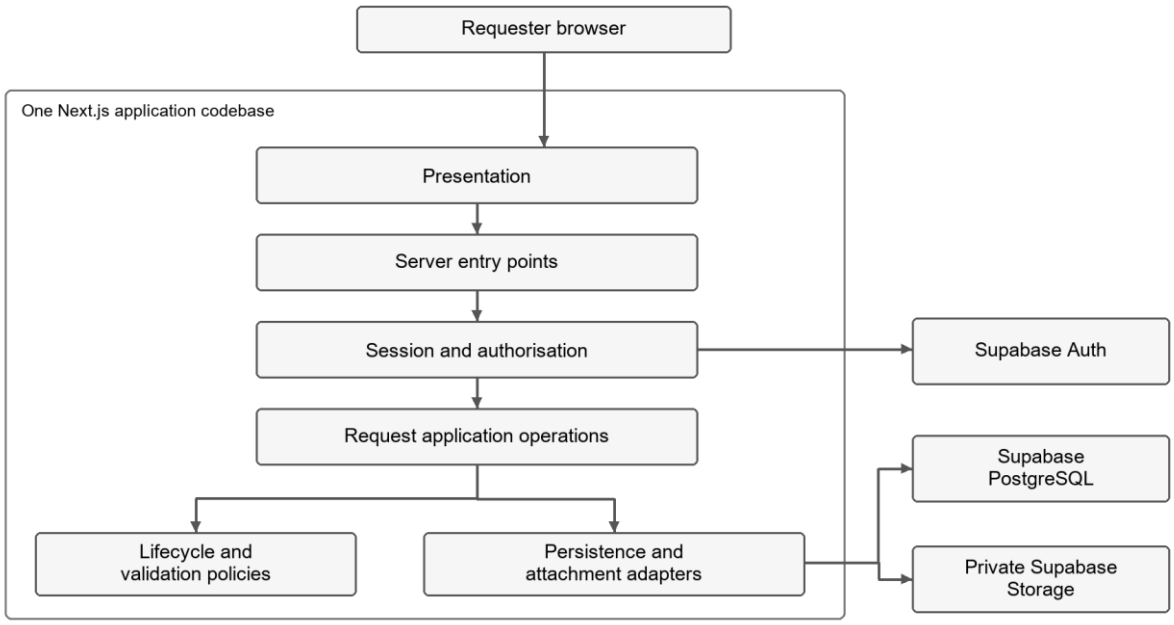


Figure 1 CivicConnect logical component architecture

14.3 Component responsibilities

Responsibility	Current location	Owner	Boundary
Pages and presentation	src/app	Ryno	Navigation, forms, feedback and readable request details
Request UI and server integration	src/features/requests	Ryno and Steven	Browser validation plus protected server functions; server remains authoritative
Request lifecycle policy	src/domain/requests	Willem	Framework-independent forward transition rules
Supabase clients	src/lib/supabase	Steven	Browser and server session clients using environment configuration
Schema and policies	supabase/migrations	Steven with Willem review	Versioned relational schema, triggers, grants and RLS
Attachment boundary	Deferred	Willem with Steven	Approved design direction; no working adapter or upload path yet

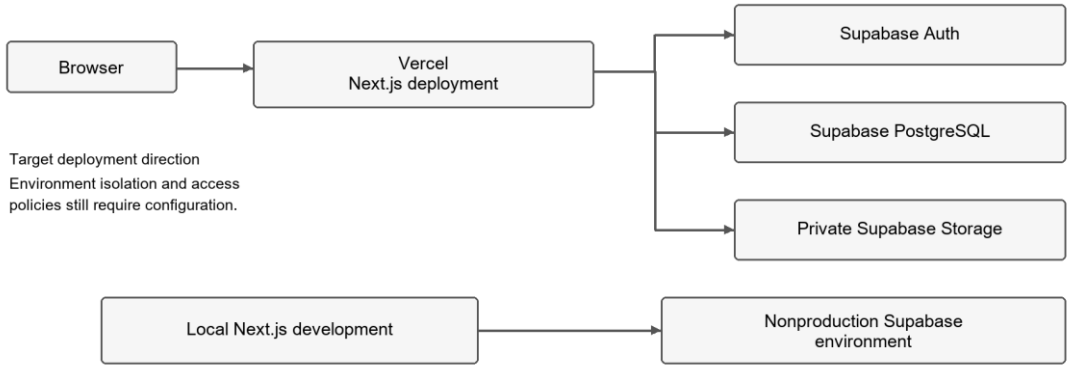


Figure 2 CivicConnect deployment direction

15 Data and persistence baseline M2 baseline snapshot

15.1 Persistence decision

ADR-002 is accepted. Supabase PostgreSQL is the primary store for structured CivicConnect data, Supabase Auth supplies authenticated identity, and private Supabase Storage is the approved direction for optional attachments. PostgreSQL matches the relational Request, Requester, Category, Status, StatusHistory and Assignment model and provides foreign keys, constraints, transactions and row-level policies. Storage objects remain a separate recovery concern and are not included in database backups.

15.2 Initial data model

Entity	Primary key	Required or significant fields	Relationship and ownership
Requester	requester_id	auth_user_id, full_name, email, created_at	One requester to many requests
Category	category_id	name	One controlled category to many requests
Status	status_id	name	One controlled status to many requests and history entries
Request	request_id	title, description, location, requester_id, category_id, status_id, sensitive_information, created_at	Belongs to one requester, category and current status
StatusHistory	status_history_id	request_id, status_id, changed_at, changed_by	Many attributable lifecycle records per request
Assignment	assignment_id	request_id, assigned_to, assigned_at	Schema foundation for later responsibility workflow

15.3 Integrity and access controls

- Required request text is non-null and cannot be blank after trimming at the database boundary.
- Category and status names are constrained to the approved seven values and request foreign keys must reference controlled rows.
- A trigger assigns Submitted before insertion and a second trigger creates the initial StatusHistory record in the same database transaction.
- The request creation time is generated by PostgreSQL. The browser does not provide requester identity, current status or creation time.
- RLS limits requesters to their mapped profile, their own requests and the related history. Broader staff and management policies remain future work.
- The current initial history trigger does not populate changed_by. Complete actor attribution remains unfinished and is retained in the RTM and risk register.

15.4 Versioned migration evidence

Migration	Purpose	Evidence status
001_initial_schema.sql	Entities, keys, constraints, indexes, seeds and initial-status trigger	Implemented in PR 24

Migration	Purpose	Evidence status
002_initial_status_history.sql	Initial history trigger	Implemented in PR 24
003_request_rls.sql	Requester-scoped grants and row-level policies	Implemented; cross-user denial test remains outstanding
004_secure_status_triggers.sql	Security-definer configuration and fixed search path for triggers	Implemented; independent security review remains later evidence

16 Technology and environment decision M2 baseline snapshot

16.1 Accepted stack

Area	Selected technology	Version or direction	Reason and current limit
Language	JavaScript	ECMAScript modules	Matches team experience; runtime validation and tests remain important
Application	Next.js App Router	16.3.7	One frontend and server application boundary
UI	React and React DOM	19.3.0	Component interface and request feedback
Runtime	Node.js and npm	Node 24.x; npm 11.x	Versions pinned by package engines and lockfile
Database and Auth	Supabase PostgreSQL and Auth	@supabase/supabase-js 2.117.2; @supabase/ssr 0.12.7	Managed relational persistence and session identity; configuration required
Attachments	Private Supabase Storage	Design direction	Approved but not implemented
Deployment	Vercel	Direction only	No hosted or production deployment claimed
Tests	Node test runner and browser checks	Node 24	Unit checks exist; database, RLS, performance and recovery suites remain incomplete

16.2 Alternatives and trade offs

ADR-001 selected JavaScript and Next.js over a separate React and Express deployment and over an ASP.NET Core alternative. One application reduces setup and cross-service coordination for the three-person team. Supabase reduces infrastructure administration but introduces provider availability, quotas, policy configuration and recovery dependencies. Vercel fits the framework but does not by itself prove environment isolation, production eligibility or operational readiness.

16.3 Configuration and deployment direction

The application reads NEXT_PUBLIC_SUPABASE_URL and NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY. Privileged keys must never use the NEXT_PUBLIC prefix or enter version control. Database changes are stored as migrations. Development, preview and production need deliberate data and credential separation. The team must verify current plan limits and recovery features before making production claims.

17 Detailed design and interfaces M2 baseline snapshot

17.1 Request transition policy

ADR-WB-01 is accepted. A central transition policy is proportionate to the small lifecycle and avoids duplicated handler rules or an unnecessary State class hierarchy. The implemented domain module is framework independent and provides canTransition and assertTransitionAllowed.

Current status	Permitted next status	Control
Submitted	Accepted or Rejected	Rejection is allowed only from Submitted
Accepted	Assigned	Skipping assignment is rejected
Assigned	In Progress	Direct resolution is rejected
In Progress	Resolved	Closure requires the Resolved state first
Resolved	Closed	Only the approved forward completion step is allowed
Closed or Rejected	None	Terminal in the current policy; reopening remains deferred

The module and its eight Node tests exist, but no database transition operation currently calls the policy. Optimistic concurrency, atomic status/history updates for later transitions and actor authorisation remain integration work. Unit tests prove the rule table only; they do not prove database consistency or RLS.

17.2 Attachment boundary

ADR-WB-02 is accepted as the future attachment design. A narrow AttachmentStore boundary and one Supabase adapter will isolate provider-specific storage calls. The application must authorise the parent request before storage access, keep the bucket private, treat object keys as identifiers rather than permissions, and clean up staged objects when database linking fails. No adapter, upload flow, file-limit policy, object backup or attachment test exists in the current baseline, so FR-001 remains partial.

17.3 Established internal interfaces

Operation	Trusted input	Outcome	Failure and security behaviour
Submit request	Authenticated session plus title, description, category ID, location and explicit sensitivity	Persisted request ID, Submitted status and server time	Server validates and returns safe field/general errors
List my requests	Authenticated session	Requester-scoped summaries	Unauthenticated or unmapped session returns no requester data
Retrieve request	Authenticated session and request ID	Permitted request fields and status history	RLS prevents other requester records; UI uses a safe unavailable result
Transition request	Current and desired status	Policy approval or controlled domain error	Database write integration and authorisation remain pending
Attachment store	Future authorised object content and metadata	Private object reference	Implementation, limits and cleanup evidence remain deferred

17.4 Validation and error handling

Browser validation improves feedback but is not authoritative. Server code trims required text, confirms a category exists and derives the requester from the authenticated session. Generic failures avoid returning SQL or credential details. A final server validation refinement is still needed because submit-request.js currently coerces sensitiveInformation with Boolean rather than rejecting every non-boolean caller value. The browser supplies a boolean, but the server boundary must remain independently correct.

18 Application and verification evidence M2 baseline snapshot

18.1 Implemented application slice

Area	Current evidence	Verified claim	Limit
Frontend	Home, login, new request, my requests and details pages	Project-specific requester workflow exists	Example route currently has an integration import defect
Submission	submit-request.js and RequestForm contract	Authenticated request insertion path exists	Server boolean validation requires refinement
Persistence	Migrations 001 to 004	Schema, seeds, triggers and requester RLS are version controlled	Fresh migration and negative policy tests were not rerun during final PED assembly
Retrieval	get-my-requests.js and getRequestDetails	Requester-scoped queries and pages exist	Cross-user denial has not been executed as a recorded test
Lifecycle	request-transition-policy.js	Approved forward transitions are executable and unit tested	Policy is not integrated into a database transition operation
Attachments	Approved ADR only	Design direction is recorded	No adapter, upload or recovery evidence

18.2 Verification record

Check	Result on 30 September 2026	Meaning
npm run lint	Passed	Current JavaScript source satisfies the configured lint rules
npm test	Passed 14 of 14	Six request-input tests and eight transition-policy tests passed
npm run build	Failed during final audit	Supabase packages were absent from the active installation and the example page imports a removed default RequestDetails component
Earlier frontend production build	Passed before PR 24 integration	Confirms Ryno's frontend contribution built before backend integration; it does not validate the current merged state
Steven's recorded Supabase demonstration	Submitted, initial history created and requester retrieved the fictional request	Contributor evidence in RTM v0.4; cross-user denial remains outstanding
Performance and recovery	Not executed	NFR-006 and NFR-007 remain Planned

The build failure is a known integration issue, not a passed verification result. Fixing the example-page import and completing a clean dependency install are required before the team demonstrates the current merged application. The documentation remains traceable by recording the failure rather than relying on the earlier pre-integration build.

18.3 Demonstrated requirement trace

NEED-001 to SCP-001 to FR-001 to RG-ASR-01, RG-ASR-03 and RG-ASR-07 to the request application responsibility to Request, Requester, Category, Status and StatusHistory to RG-ADR-01, ADR-001 and ADR-002 to migrations and submit-request.js to the recorded Supabase submission and retrieval evidence. Attachments and final negative-security verification remain open, so FR-001 is In Development rather than complete.

19 Requirements traceability matrix

This matrix continues the M1 and M2 trace. Section 19.1 preserves stakeholder, need, scope, priority, acceptance and ASR links while updating implemented data responsibilities. Section 19.2 records final local and selected hosted evidence. The verified application is commit 374c21e7c2b3d9ecc01add9ec2b957ca53139a97, with matching source hashes in F15 and successful CI13. Demonstrated status applies only to recorded scenarios; Partial retains the stated acceptance or operational limits.

Evidence key F15 means Milestone_3/Verification_2026-10-15_Final/. E15 is the earlier 15 October record. CI13 passed for the final application commit 374c21e; CI11 remains historical evidence for the earlier CR-M3-01 commit 5a4e4f4. F15 contains 55 local functional cases, selected hosted checks, performance and local recovery reports. Full component paths appear in section 24.1. RG-ASR-01 to RG-ASR-07 retain the M2 driver identifiers.

19.1 Requirement source architecture and data trace

Requirement	Source stakeholder need scope priority	Acceptance	ASR	Architecture responsibility	Data and persistence
FR-001 Submit request	S4; STK-001/002; NEED-001; SCP-001; Must	AC1 to AC6	RG-ASR-01/03/07	Request submission path	Request, Requester, Category, Status, History
FR-002 Controlled categories	S4; STK-001/004; NEED-002; SCP-002; Must	AC1 to AC5	RG-ASR-03/07	Category reference data	Seven seeded categories and Request FK
FR-003 View current status	S4; STK-001; NEED-003; SCP-003; Must	AC1 to AC4	RG-ASR-01/07	Requester status/detail	Request status FK and RLS
FR-004 View history or list	S4; STK-001; NEED-003; SCP-004; Must	AC1 to AC4	RG-ASR-01/05	Requester list	Requester to Request; retention open
FR-005 In-app feedback	S4; STK-001; NEED-003; SCP-005; Must	AC1 to AC6	RG-ASR-07	Feedback presentation	Persisted status and outcome
FR-006 Staff work view	S4; STK-002; NEED-004/009; SCP-006; Must	AC1 to AC4	RG-ASR-01/02	Staff work queue	Service membership, category scopes, sensitivity and RLS
FR-007 Search filter sort	S4; STK-002; NEED-004; SCP-007; Must	AC1 to AC5	RG-ASR-04/07	Operational query layer	Scoped lists; browser search, filter and sort; indexes
FR-008 Authorised details	S4; STK-002; NEED-004/009; SCP-008; Must	AC1 to AC4	RG-ASR-01/02	Request detail and access	Sensitive flag and requester RLS
FR-009 Assignment	S4; STK-002/004; NEED-005; SCP-009; Must	AC1 to AC5 retained; AC6 added by CR-M3-01, 15 October 2026	RG-ASR-01/03	Assignment service	Assignments with actor, reason, visibility and version guard
FR-010 Status transitions	S4; STK-002/004; NEED-006; SCP-010; Must	AC1 to AC7	RG-ASR-01/03	Lifecycle policy	Statuses, attributable history and atomic workflow RPC
FR-011 Actions comments resolution	S4; STK-002; NEED-006; SCP-011; Must	AC1 to AC5	RG-ASR-03	Request activity	Comments with author, time, visibility and resolution feedback
FR-012 Resolve or close	S4; STK-002/004; NEED-006; SCP-012; Must	AC1 to AC5	RG-ASR-01/03	Resolution and closure	Authorised resolution and manager closure with history
FR-013 Service activity	S4; STK-003; NEED-007; SCP-013; Must	AC1 to AC4	RG-ASR-01/07	Management activity	Scoped requests and derived service activity
FR-014 Open and overdue	S4; STK-003; NEED-007; SCP-014; Must	AC1 to AC5	RG-ASR-04/07	Overdue calculation	Request target_at, status and overdue calculation
FR-015 Analysis	S4; STK-003; NEED-008; SCP-015; Must	AC1 to AC5	RG-ASR-01/04	Reporting and analysis	Scoped category and status aggregates
FR-016 Accountability	S4; STK-003; NEED-008; SCP-016; Must	AC1 to AC5	RG-ASR-03	History and accountability	Status history, assignments, actors and protected reason history
NFR-001 Least privilege	S4; STK-001/002/003/005; NEED-009; SCP-017; Should	AC1 to AC5	RG-ASR-01	Authentication and access	Auth mapping and requester RLS

Requirement	Source stakeholder need scope priority	Acceptance	ASR	Architecture responsibility	Data and persistence
NFR-002 Sensitive protection	S4; STK-001/002/005; NEED-009; SCP-017; Should	AC1 to AC5	RG-ASR-02	Sensitive data access	Sensitive flag and requester RLS
NFR-003 Auditability	S4; STK-002/003/006; NEED-006/008; SCP-010/011/016; Should	AC1 to AC4	RG-ASR-03	Lifecycle audit trail	Attributable lifecycle, comments and assignment history
NFR-004 Usability	S4; STK-001/002/003; NEED-001/003/004; multiple SCP; Should	AC1 to AC5	RG-ASR-07	Requester UI and feedback	Readable status and request views
NFR-005 Integrity	S4; STK-002/003; NEED-001/005/006/008; multiple SCP; Should	AC1 to AC5	RG-ASR-03	Relational integrity	PK FK checks triggers
NFR-006 Performance	S4; STK-003/005/006; NEED-010; SCP-018; Should	AC1 to AC4	RG-ASR-04	Performance responsibility	Measured local and hosted operations; 1000-row local list exercise
NFR-007 Recovery	S4; STK-005/006; NEED-010; SCP-018; Should	AC1 to AC4	RG-ASR-05	Backup and restore	Coordinated local database, object and configuration snapshot
NFR-008 Maintainability	S4; STK-006; NEED-010/011; SCP-018; Should	AC1 to AC5	RG-ASR-06	Repository and traceability	Versioned code migrations ADRs and RTM

19.2 Design technology implementation and verification trace

Req	Design or interface	Technology	Implementation evidence	Test expectation result and execution evidence	Status	ADR change or risk
FR-001	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	request-form.js; validate-request.js; submit-request.js; workflow/service.js; schema DB01, DB12, E2E01; validation unit cases	Validation unit; DB01/11/12; E2E01. Expected: Reject invalid fields; save Submitted, timestamp and optional attachment. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Saved request, initial state/time and private attachment verified locally and against hosted Supabase. Contact identity follows the session/profile model. R11 R13.
FR-002	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Controlled category selection and reference data Validation unit cases; DB01; E2E01 Maintenance selection	Validation unit; DB01; E2E01. Expected: Reject uncontrolled category; retain selected controlled category. Actual: pass for listed cases. F15; CI13.	Partial	Partial: controlled selection demonstrated; the browser case does not independently enumerate all seven baseline categories. ADR-002
FR-003	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	StatusBadge and protected request detail E2E01, E2E02, DB02	DB02; E2E01/02. Expected: Display stored state; deny another requester. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Demonstrated: own recorded status and denial of other-requester access. R3 R14
FR-004	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Own-request list and detail/history E2E01; persisted history in DB01/DB04	DB01/04; E2E01. Expected: Retrieve owned requests and recorded history. Actual: pass for listed cases. F15; CI13.	Partial	Partial: retrieval and history demonstrated; final retention period remains unconfirmed. Q-006

Req	Design or interface	Technology	Implementation evidence	Test expectation result and execution evidence	Status	ADR change or risk
FR-005	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Request detail status and public history/comments E2E02; transition-policy unit cases	Transition unit; E2E02. Expected: Show permitted event/status feedback. Actual: pass for listed cases. F15; CI13.	Partial	Partial: updates, resolution and closure visible; every accepted/rejected feedback path is not separately demonstrated by E2E02. Q-005
FR-006	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Scoped staff work service and database policy DB03, DB08, DB09; E2E02	DB03/08/09; E2E02. Expected: Return only permitted work; enforce sensitivity. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Demonstrated: permitted staff work and category/clearance restrictions. Q-002 R3
FR-007	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Request/work lists and scoped query/filter controls Responsive UI case; DB scope cases	Responsive case; scope integration. Expected: Apply selected criteria within permitted data. Actual: pass for listed cases. F15; CI13. Hosted search/filter/sort interactions timed after loading; full criterion and 1000-row hosted coverage remain incomplete.	Partial	Partial: interface filtering and scoped retrieval exist; complete staff search/filter/sort acceptance is not established by the recorded browser case. Q-001 R9
FR-008	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Protected request detail and access policies DB02, DB03; E2E01; API01/API03	DB02/03; API01/03; E2E01. Expected: Return authorised details; conceal protected data. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Demonstrated: role/scope, sensitive clearance, requester isolation and generic failure. Q-002 Q-006
FR-009	Workflow and protected UI/API; assignment-history RPC; CR-M3-01	JS; Next.js; PostgreSQL; ADR-001/002	WF service; SQL005/007/008; assignment form and history. CR01-03; CRDB01-07; CRE2E01.	CR01-03; CRDB01-07; CRE2E01. Expected: Require reason for changed assignee; retain attributable history. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	CR-M3-01 AC6 validated locally and in hosted manager reassignment; reason, actor, timestamp and visibility persist. Migration 008 preserves assignment data and permissions. R11 R12.
FR-010	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Central transition policy; workflow RPC; migration 005 Transition unit cases; DB01/04/05/08/10; E2E02	Transition unit; DB01/04/05/08/10; E2E02. Expected: Permit policy transitions; reject skipped/unauthorised changes. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Demonstrated: controlled sequence, rejection/reopening rules, attribution and blocked invalid operations. D8 R11 R12 R15
FR-011	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Staff activity/comment operation; visibility policy DB07; DB04; E2E02	DB07; CRDB06; E2E02. Expected: Retain notes and resolution with visibility control. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Demonstrated: attributable resolution notes and requester-hidden private comments. Requester comment creation is not required. Q-005
FR-012	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Resolution/closure operations and policy DB04, DB08; E2E02	DB04/08; E2E02. Expected: Authorised resolution/closure with actor attribution. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Demonstrated: permitted resolution and manager closure with attribution; requester mutation denied. D8 Q-003

Req	Design or interface	Technology	Implementation evidence	Test expectation result and execution evidence	Status	ADR change or risk
FR-013	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Scoped management data and recorded lifecycle history DB04; E2E02 management page	DB04; E2E02. Expected: Show scoped recorded service activity. Actual: pass for listed cases. F15; CI13.	Partial	Partial: scoped activity exists; the browser assertion verifies category display rather than every representative activity criterion. Permission model
FR-014	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	reporting.js overdue rules and workflow target BB01, BB02; scope/history integration cases	BB01/02. Expected: Correct open/terminal and overdue boundary classification. Actual: pass for listed cases. F15; CI13.	Partial	Partial: deadline boundary and terminal exclusion demonstrated; target-date setting is implemented but not established by those two black-box cases alone. Q-004
FR-015	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Scoped management aggregates and reporting.js BB03; E2E02; DB03	BB03; E2E02. Expected: Correct scoped category/status aggregates. Actual: pass for listed cases. F15; CI13.	Demonstrated locally	Demonstrated for supplied category/status scenarios; permission scope is enforced by queries. No predictive analytics claim. R9
FR-016	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; ADR-001/002	Assignment, history attribution and management/detail views DB04; E2E02	DB04; CRDB03; E2E02/CRE2E01. Expected: Reconstruct progression and responsibility. Actual: pass for listed cases. F15; CI13.	Partial	Partial: retained progression and responsibility support accountability; these cases do not establish all management interpretation criteria. R11 R12 CR-M3-01.
NFR-001	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	Protected entry points and database least-privilege rules DB02/03/08/09; API01/02/04	DB02/03/08/09; CRDB05; API01/02/04. Expected: Deny unauthorised role/scope and mutation access. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Selected hosted Auth, requester isolation, scoped staff access and protected Next.js routes passed. This is bounded verification, not a complete deployment/security certification. R3 R14 R15.
NFR-002	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	Sensitivity, private comments and attachment controls DB02/03/07/11/13; E2E01; BB05	DB02/03/07/11/13; CRDB06; E2E01. Expected: Protect sensitive/internal information and attachments. Actual: pass for listed cases. F15; CI13.	Partial	Selected hosted private attachment, sensitive-data and internal comment/reason restrictions passed. Retention remains unconfirmed; full operational security is deferred. R3 R14.
NFR-003	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	Attributable history, assignment and activity records DB01, DB04, DB07; E2E02	DB01/04; CRDB03; E2E02. Expected: Attribute lifecycle and responsibility changes. Actual: pass for listed cases. F15; CI13.	Demonstrated in recorded local and hosted scenarios	Demonstrated for recorded lifecycle and activity scenarios; no external audit certification claim. R11 CR-M3-01.
NFR-004	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	Task feedback, responsive shell and keyboard behavior E2E01/E2E02; responsive UI case	E2E01/02; CRE2E01; responsive case. Expected: Complete defined tasks with understandable feedback. Actual: pass for listed cases. F15; CI13.	Partial	Partial: defined local tasks support usability; no comprehensive user study or every oversight interpretation criterion established. R6

Req	Design or interface	Technology	Implementation evidence	Test expectation result and execution evidence	Status	ADR change or risk
NFR-005	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	Atomic workflow; expected-version guard; migration 008 PT409; service conflict mapping. DB05/06/12; CRDB02/04/05/08; API05; hosted focused conflict report.	DB06/CRDB08/API05 and hosted check: reject stale write with 409 and reload message; preserve saved state; current version succeeds. F15. One post-load timeout is retained as HOSTED-TIMEOUT-01.	Demonstrated locally	Demonstrated: rollback, stale updates, boundary validation and truthful cleanup outcomes. R11 R12 R15 CR-M3-01.
NFR-006	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	Concurrent independent detail reads after parent authorisation; local-performance.json and hosted-performance.json in F15.	Local 1000-row/3-identity list p95 72.15 ms. Hosted small-data p95: list 1232.54 ms; management 1877.22 ms; details 1473.70 ms; submission 1930.99 ms; UI controls below 46 ms. F15 section 28.	Partial	Partial: measured operations meet their targets under the stated workloads. Not every operation is verified at 1000 hosted rows; sustained capacity and latency variability remain M4 work. R8 R9.
NFR-007	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	scripts/simulation-backup.mjs; scripts/test-recovery.mjs	Local 1000-request coordinated restore 468.60 ms; simulated loss 507 ms. Contents, objects, access boundaries and continued workflow passed. F15 local-recovery.json.	Partial	Controlled local rehearsal passed. Routine scheduling, live mutation backup and hosted database/Storage/Auth restore remain unverified and scheduled for final submission. R8 R16.
NFR-008	Workflow and protected UI/API; established M2 design extended	JS; Next.js; PostgreSQL; M2 quality driver	Stable IDs; migrations 001-008; PR30-33; earlier CI11; final local source and report hashes in F15.	CI11; CRDB07; PR evidence. Expected: Stable trace, versioned change, approvals and passing checks. Actual: pass for listed cases. F15; CI13.	Partial for final source baseline	Final local gates passed; source hashes identify the tested checkout. Earlier approvals and CI are retained as history; a new remote review/CI result for later corrections is not asserted. R2 R5.

20 Risk assumption and dependency registers

20.1 Current risk register

ID	Risk	P I Priority	Mitigation	Contingency	Owner	Status
R1	Scope creep after baseline	M/H/H	CR-M3-01 records a bounded request and impact analysis; no external ingestion or predictive scope added.	Reprioritise approved scope	Willem	Open; mitigated within tested scope
R2	GitHub governance not enforced	M/H/H	Branch protection confirmed enforced by Ryno. PRs 30-32 have two GitHub approvals each; PR33 team approvals confirmed and CI11 passed.	Stop non-compliant merges	Ryno	Open; mitigated within tested scope
R3	Auth or policy gaps expose request data	M/H/H	Local and selected hosted role/scope/sensitivity denials passed, including requester isolation and internal reason protection. Full hosted assurance remains a final-submission task.	Disable path and repair policy	Steven	Open; mitigated within tested scope
R4	Design and implementation drift	M/M/M	Current component map and RTM linked to candidate; architecture ADRs retained.	Reopen decision and correct evidence	Willem	Open; mitigated within tested scope
R5	Weak PED control or status confusion	L/H/M	Continued PED and dated source/report hashes distinguish historical CI from later local verification; final submission must identify the committed source.	Correct baseline and repeat review	All	Open; mitigated within tested scope
R6	Schedule and learning burden	M/H/H	Small local release scope and repeatable setup reduce coordination; remaining release work retains schedule exposure.	Resequence later features	Ryno	Open; mitigated within tested scope
R7	Team governance breakdown	M/M/M	Two-reviewer policy retained; contribution approvals and CR-M3-01 joint decision recorded.	Escalate unresolved breakdown	All	Open; mitigated within tested scope
R8	Environment or recovery mismatch	M/H/H	Local coordinated restore passed; selected hosted integration and timing exercised. Routine backup scheduling and hosted restore remain final-submission work.	Restore known-good isolated environment	Willem	Open; mitigated within tested scope
R9	Provider quota outage cost or dependency failure	M/H/H	Production audit is clean; dated development dependency findings remain. HOSTED-TIMEOUT-01 records one post-workload timeout with successful targeted retests.	Reviewed retry or alternative	All	Open; mitigated within tested scope

ID	Risk	P I Priority	Mitigation	Contingency	Owner	Status
R10	Incorrect or weakly verified AI output	L/H/H	AI register, source review, peer review and real regression/CI results; selector error corrected.	Correct or withdraw material	All	Open; mitigated within tested scope
R11	Request and history become inconsistent	M/H/H	Transactional history and reason writes; DB12 and CRDB02/04 protect rollback consistency.	Block mutation and investigate	Willem	Open; mitigated within tested scope
R12	Concurrent updates overwrite state	M/H/H	Migration 008 uses non-retryable PT409 for stale business versions. DB06, CRDB08, API05 and hosted checks verify prompt conflict and unchanged data; transient timeout retained.	Reload and require fresh action	Willem	Open; mitigated within tested scope
R13	Attachment upload leaves orphan	M/M/M	ATT01/02 cover cleanup and truthful failure reporting; storage failures retain orphan risk.	Retry verified cleanup	Willem	Open; mitigated within tested scope
R14	Storage or privileged credentials bypass access	M/H/H	Hosted private uploads/downloads and access denial passed. Internal assignment reasons use a protected RPC; privileged credential handling and recovery remain operational concerns.	Revoke credentials and repair policy	Steven	Open; mitigated within tested scope
R15	Direct writes bypass lifecycle policy	M/H/H	Direct mutation denied; lifecycle/reassignment rules independently enforced in SQL.	Disable bypass and review records	Willem	Open; mitigated within tested scope
R16	Database backup omits attachment objects	M/H/H	Local backup includes database, object bytes and configuration; hosted coordinated recovery remains unverified.	Restore consistent set and restrict gaps	Willem	Open; mitigated within tested scope

20.2 Assumptions and dependencies

ID	Assumption or dependency	Current validation and consequence
ASM-001	One organisation is the initial context	Retained; multi-tenant design not committed
ASM-002	In-app feedback is the initial mechanism	Retained; external delivery remains deferred
ASM-003	Coordination is a responsibility set	Retained; validate with organisation evidence later
ASM-004	Four hours per member daily includes review	Confirmed planning basis
ASM-005	Fictional non-sensitive data is used	Retained for development and demonstration

ID	Assumption or dependency	Current validation and consequence
ASM-006	Optimistic concurrency is proportionate	Version tokens, stale rejection and concurrent workflow checks are implemented; PT409 avoids the observed hosted retry issue. Real contention capacity remains a limit.
ASM-007	Selected stack can support the first slice	55 local cases, production builds and selected hosted journeys passed; no current build failure is asserted.
ASM-008	Auth identity can map to Requester and history actors	Auth identity, requester profiles and lifecycle/assignment actors are implemented and verified in recorded local and hosted cases.
ASM-009	Development preview and production can be separated	Local simulation and production build with hosted test provider were kept separate; a public production topology is not claimed.
DEP-M2-01	Supabase and Vercel availability and limits	External dependency; verify account and plan evidence
DEP-M2-02	Schema grants and policies enforce the write boundary	RLS, grants and negative/bypass tests passed in the local suite; selected hosted isolation checks also passed.
DEP-M2-03	Attachments need limits retention cleanup and object backup	Private attachments, size/type checks, cleanup tests and local coordinated restore are implemented. Retention, daily scheduling and hosted recovery remain final-submission work.

21 Forward engineering baseline

Concern	Why it matters	Current evidence and final-submission follow-up	Risk links
Security	Auth, server checks, RLS and private storage must agree	Selected hosted access, private Storage and local policy tests passed; broader threat/security and retention validation remain.	R3 R14 R15
Testability	Unit rules do not prove database transactions or provider policies	55 local functional cases plus selected hosted journeys and negative checks passed; extend acceptance coverage where partial.	R4 R11 R12
Deployment	Framework and managed services require repeatable configuration	Simulation and hosted-provider production builds passed; commit 374c21e and CI13 identify the reviewed application baseline.	R6 R8 R9
Migration and recovery	Schema and object references affect rollback and restore	Migration 008 upgrade and coordinated local restore passed; routine scheduling and hosted restore remain for final submission.	R8 R16
Observability	Failed writes and cleanup need safe diagnosis	Timeout and failure evidence retained; production monitoring, correlation and alerting remain for final submission.	R11 R13
Maintainability	Central policy, schema and RTM must remain aligned	RTM, corrections, tests and source hashes updated; existing architecture decisions retained.	R4 R5
Operational cost	Provider quotas and storage growth affect feasibility	Production dependencies audited; hosted quota, growth and operational-cost review remain final-submission work.	R9

22 Engineering decision log

ID	Decision	Status	Consequence
D1	Initial web delivery excludes native mobile applications	Accepted	Reassess only through controlled scope change
D2	Protected main and two non-author approvals	Accepted	Every substantive author depends on both reviewers
D3	One evolving PED is the controlled baseline	Accepted	M1 history is preserved in v2.0
D4	JavaScript Next.js Supabase and Vercel stack	Resolved by ADR-001	Environment and deployment evidence remain separate
D5	node:test with browser verification as initial direction	Implemented	24 unit, 22 database and 9 browser/API cases passed locally; CI13 passed for the final application.
D6	Evidence-based negotiation for stakeholder conflicts	Accepted	Unresolved policy remains traceable
D7	AI assistance allowed with disclosure and verification	Accepted	Each member remains responsible for the submitted work
D8	Central request transition policy	Accepted ADR-WB-01; integrated	Atomic database workflow, attribution and invalid-transition tests.
D9	Narrow AttachmentStore with one Supabase adapter	Accepted ADR-WB-02; implemented	Private attachment adapter, cleanup tests and local coordinated restore.
D10	Internal request contracts evolve with callers	Implemented and extended	Protected internal contracts; m3_assignment_history adds visibility-safe reason history under CR-M3-01.
D11	Modular monolith architecture	Accepted RG-ADR-01; conformance evidenced	Single Next.js release; actual component map in section 24.
D12	Supabase PostgreSQL persistence	Accepted ADR-002; extended	Migrations 001-008; transactions, protected history, local recovery and selected hosted checks verified.
D13	Local simulation staging on port 3103	Selected 13 October 2026	Separate production build with fictional local provider; hosted/operational limits retained.
D14	Require reasons for changed assignees under CR-M3-01	Jointly approved 15 October 2026	Original CR-M3-01 baseline passed CI11 and 53 local cases; final suite passed 55 local cases, selected hosted checks and CI13.
D15	Use PT409 for business version conflicts	Implemented and locally/hosted verified, 15 October	Migration 008 preserves the existing function and permissions; service returns a reload message. Avoid retryable 40001 for this business conflict.
D16	Fetch independent detail information concurrently	Implemented and measured, 15 October	Authorise parent first; retain all policies. Hosted detail p95 improved from 2415.75 ms to 1473.70 ms. No new service or dependency.

23 AI usage and baseline approval

23.1 Milestone 2 AI usage entries

Date	Student	Tool	Engineering task	Human verification	Decision and issues
30 Sep	Ryno Goetz	OpenAI Codex	Architecture, frontend and PED integration	Compared with the Master and M2 briefs, PED v1.0, repository evidence, tests and rendered Word output	Accepted with corrections; unsupported completion claims retained as open
30 Sep	Steven Riaan Piek	ChatGPT	Requirements, data, persistence, implementation and RTM support	Compared with approved requirements, schema/code evidence and the recorded Supabase demonstration	Accepted with corrections; incomplete security and verification retained
30 Sep	Willem Booysen	OpenAI Codex	Detailed design, register updates and transition-policy support	Compared with M1 requirements, Assignment 2 research, repository code and unit-test results	Accepted with corrections; attachment and DB-transition evidence remain deferred

23.2 Milestone 2 approval

All three members approved the M2 architecture, technology, persistence and detailed design decisions on 30 September 2026. Merged PRs 17 to 25 form the controlled repository evidence for the integrated contribution set. Approval confirms the baseline and its recorded limitations; it does not convert Planned or In Development requirements into completed product acceptance.

Member	M2 baseline status	Date
Ryno Goetz	Approved	30 September 2026
Steven Riaan Piek	Approved	30 September 2026
Willem Booysen	Approved	30 September 2026

23.3 M2 carry-forward concerns and M3 disposition

Server boolean validation and meaningful submission tests were implemented and verified in the M3 suite.

Cross-user denial tests and scoped staff/management policies passed locally; selected hosted requester isolation, sensitive access and private Storage cases also passed. Broader operational limits remain in section 29.

Transactional lifecycle operations, actor attribution and stale-version checks were integrated and verified.

Private attachments, validation, cleanup behavior and coordinated local backup/restore were implemented and tested.

The local performance and timed recovery exercises passed. Their operation, environment and scheduling limits remain in section 28.

24 Milestone 3 integrated application baseline

The M3 application integrates requester submission/retrieval, scoped staff work, management oversight, private attachments, lifecycle control and CR-M3-01. The architecture remains the accepted modular monolith. The selected presentation staging remains local simulation on port 3103; additional production-build checks used hosted Supabase on local port 3104. Section 32 identifies application commit 374c21e, successful CI13 and the source-hash manifest, including migration 008 and the detail-read correction.

24.1 Architecture conformance

One Next.js application contains presentation, protected entry points, shared workflow operations and domain rules. Supabase PostgreSQL, Auth and private Storage remain the hosted technology direction. Local verification substitutes PGlite with the real migrations and policies, a loopback provider and temporary object storage. No separate backend deployment was introduced. M2 diagrams in section 14 describe the retained architecture direction; the table below identifies actual M3 responsibilities.

Responsibility	Actual implementation	Boundary
Presentation	src/app; src/components; src/features/workflow	Shared role-aware screens and task feedback
Protected operations	src/app/api; src/features/requests/submit-request.js	Session identity, validation and origin checks
Workflow service	src/features/workflow/service.js	Scoped retrieval, submission and coordinated changes
Domain policy	src/domain/requests/request-transition-policy.js; assignment-reason.js; reporting.js	Lifecycle, reason validation and reporting rules
Attachment adapter	src/features/attachments/attachment-store.js	Inspection, private object operations and cleanup
Persistence	supabase/migrations/001 through 008	Integrity, RLS, atomic changes and protected reason history

24.2 Interface and implemented behavior

The interface retains the approved green palette, role-aware workspace shell, responsive request lists, readable forms and request history. Requesters can sign in, validate and submit a request, retrieve their own records and attach one permitted private PDF, PNG or JPEG up to 5 MiB. Staff work within category and sensitive-clearance rules. Managers use scoped assignment, closure and reporting operations. A changed assignee requires a reason, with saved assignment history and field-associated validation feedback. Brief animations respect reduced-motion preferences.

Requester-visible updates are distinguished from internal comments and internal assignment reasons. Requesters do not acquire staff mutation rights through the interface. Server and database checks enforce permissions independently of button visibility. Empty, validation, denied-access and failure states are represented in the supported workflows.

24.3 Runtime configuration and migrations

The application declares version 0.3.0, JavaScript modules, Next.js 16.3.8, React 19.3.0, Node.js 24 and npm 11. Use npm ci. A fresh database requires migrations 001 through 008 in order. Migration 005 adds workflow, 006 private attachments, 007 reassignment reasons

and protected history, and 008 returns PT409 for an outdated version without changing the workflow or existing permissions. Hosted verification confirmed the new conflict response after operator application.

Hosted configuration uses NEXT_PUBLIC_SUPABASE_URL and NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY in the ignored .env.local. These are public client settings; privileged credentials do not belong in public variables or Git. Simulation scripts provide a clearly fictional loopback configuration. Build-time settings require a rebuild when switching environments. Hosted Auth UUIDs must match provisioned profiles; SQL example IDs are not real accounts.

24.4 Local staging and demonstration access

From the checked-out candidate, the repeatable local sequence is:

npm ci

npx playwright install chromium

npm run build:simulation

npm run staging:simulation

Open http://127.0.0.1:3103 and keep the terminal running. The local provider uses 127.0.0.1:54329. Ctrl+C stops manual staging. Stop manual staging before browser/performance tools that start the same ports. Ordinary npm run dev uses development configuration and does not start this selected simulation. The loopback URL is accessible on the demonstration machine; the commands and fixture accounts provide the assessor-accessible equivalent.

Fictional account	Role and demonstration purpose
requester@example.test	Submission and owned-request retrieval
other@example.test	Cross-requester denial
staff@example.test	Maintenance staff with sensitive clearance
manager@example.test	Maintenance manager without sensitive clearance
restricted@example.test	Maintenance staff without sensitive clearance
admin@example.test	Administration without automatic business access

All local fixtures use Fictional-test-password-1!. Staff and manager fixtures have no requester profile, so submission uses a requester fixture. These accounts contain fictional data and do not identify hosted users.

24.5 Environment differences and recovery boundary

The simulation runs the application and real migrations/policies, with temporary data and objects. Additional hosted tests exercised real Auth, PostgreSQL/PostgREST and private Storage; these do not establish full environment equivalence, operational availability or hosted recovery. A coordinated local snapshot was restored successfully, but ordinary staging does not schedule or restore backups automatically. Hosted recovery, monitoring and larger workloads remain final-submission work under the retained architecture and risk controls.

24.6 Additional hosted verification environment

A separate production Next.js build on 127.0.0.1:3104 used the configured hosted Supabase project. Hosted public URL/key settings stayed in ignored .env.local; privileged credentials were not supplied to the application or published in this PED. Build with npm run build, then run npm start -- --hostname 127.0.0.1 --port 3104. Rebuild when switching back to the simulation. This loopback application with hosted services is a verified test arrangement, not a public production deployment.

Hosted requester@example.test and other@example.test had requester profiles. staff@example.test and restricted@example.test were scoped to IT support without sensitive clearance; manager@example.test had all seven category scopes and sensitive clearance. These differ from the Maintenance simulation fixtures in section 24.4. Ten hosted browser checks passed; migration 008 also passed focused conflict checks. Fictional records and attachments were used throughout.

25 Persistence traceability and acceptance coverage

25.1 Data and application contracts

The schema retains Requester, Request, Category, Status, StatusHistory and Assignment and adds scoped service membership, request comments, reopening requests and private attachment controls. Session-derived identity is authoritative. Creation records Submitted, creation time and initial attributable history in one database transaction. Controlled workflow operations apply lifecycle, scope and expected-version checks. Failure does not silently leave request state and history inconsistent.

Contract	Outcome and control
Submission	Validate required input and category; derive requester from session; persist and return confirmation.
Own list and details	Return permitted data; deny cross-requester access with a generic unavailable response.
Staff and manager work	Enforce permitted category and sensitive clearance before exposing work and reporting.
Workflow change	Validate operation, actor and expected version; record attributable history atomically.
Assignment history	m3_assignment_history exposes permitted metadata/reasons; internal reasons remain hidden from requesters.
Attachment	Inspect size/type/signature; authorise parent request; private download and truthful cleanup failure handling.

25.2 Evolved RTM and controlled criteria

Section 19 is the current RTM. It retains all 24 requirement identifiers and original criteria, with FR-009 AC6 added through CR-M3-01. F15 resolves to final local and selected hosted evidence; CI13 passed for application commit 374c21e, with CI11 retained as history. Passing selected cases does not establish every acceptance criterion. The two verification corrections add no new business capability or acceptance waiver.

25.3 Representative requirement to evidence trace

NEED-001 and SCP-001 lead to FR-001 and its submission criteria. The request form and validate-request.js collect and validate input; submit-request.js invokes the workflow service using the session identity; database insertion creates the request and initial history. DB01 checks time/state/actor, DB12 checks failed-history rollback, and E2E01 checks browser validation, private attachment and requester retrieval. The current unit, database and browser reports are indexed in section 32.

FR-009 follows NEED-005 and SCP-009 into authorised responsibility changes. CR-M3-01 adds the required reassignment reason. assignment-reason.js, the workflow service and SQL007 implement the rule; CR01-03 and CRDB01-07 verify boundaries, atomicity, privacy, permission and upgrade behavior; CRE2E01 verifies manager reassignment and visible persisted history after reload.

25.4 Acceptance limits

The agreed core has integrated local and selected hosted evidence. The retention period and some broader oversight/acceptance criteria remain incomplete. NFR-006 has one 1000-row local read exercise and additional small-data hosted measurements, not every operation at the full baseline workload. NFR-007 has a controlled local restore, not routine scheduled or hosted recovery. Ryno scheduled these remaining quality and operational concerns for the final submission; the current targets remain unchanged.

26 Quality strategy and verification evidence

Verification prioritises submission integrity, least privilege, sensitive/internal information, valid lifecycle changes, stale updates and the requester-to-staff-to-manager journey. Unit tests isolate rules; PostgreSQL integration tests exercise policies and transactions; browser/API tests exercise protected boundaries and full journeys. Boundary value analysis, equivalence partitioning, decision tables and state-transition testing are applied to relevant failure modes.

26.1 Current execution and meaningful coverage

Final 15 October verification passed 24 unit tests, 22 database integration tests and 9 browser/API cases: 55 unique local cases. The browser/API suite includes three journeys, five API cases and one responsive case. BB01 to BB05 overlap legitimately with the unit suite. The additional ten hosted browser checks and focused conflict checks are separate provider-validation evidence, not an inflation of the 55-case total. Performance and recovery exercises are also separate. No code-coverage percentage is asserted.

Check	Recorded result	Controlled evidence
Unit	24 passed	Unit report
Database integration	22 passed	Database report
Browser and API	9 passed	Browser/API report
Lint and secret scan	Final local checks and CI13 passed; CI11 retained as history	Final verification index
Simulation build	Simulation and hosted-provider production builds passed locally	Final verification index
Static UI audit	Zero findings	UI audit
Production dependencies	Zero reported vulnerabilities	Production audit
Hosted browser	10 selected checks passed	Hosted browser results
Hosted stale-version correction	Prompt 409, helpful message, unchanged data and fresh update passed	Focused conflict results

26.2 Selected test catalogue

The following selected cases were executed successfully on 15 October against fictional fixtures. They identify basis, technique, input and expected/actual outcome. Rule cases use tests/m3-rules.test.js; DB cases use tests/integration/database.test.js; journeys use tests/e2e/workflow.spec.js. The full dated reports preserve all executed case names. Additional CR cases are traced in section 30.

26.2.1 BB01 FR-014

Basis and selection: FR-014. Prevent early or missed overdue classification. Technique: Boundary value analysis.

Precondition and input: Before, exactly at and after overdue deadline. Expected outcome: Only after the deadline is overdue. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/unit.xml.

26.2.2 BB02 FR-014

Basis and selection: FR-014. Avoid reporting terminal work as overdue. Technique: Decision table.

Precondition and input: Terminal statuses and absent deadline. Expected outcome: Terminal requests and missing deadlines are not overdue. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/unit.xml.

26.2.3 BB03 FR-015

Basis and selection: FR-015. Protect management summary accuracy. Technique: Equivalence partitioning.

Precondition and input: Scoped Submitted, Closed and Resolved requests. Expected outcome: Correct scoped aggregate totals. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/unit.xml.

26.2.4 BB04 FR-001 and NFR-005

Basis and selection: FR-001 / NFR-005. Bound uploads and reject unusable files. Technique: Boundary value analysis.

Precondition and input: Empty, exactly 5 MiB and oversized attachment. Expected outcome: Reject empty/oversized; permit valid maximum size. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/unit.xml.

26.2.5 BB05 NFR-002 and NFR-005

Basis and selection: NFR-002 / NFR-005. Reject misleading attachment declarations. Technique: Equivalence partitioning.

Precondition and input: Spoofed PDF, mismatched MIME and valid JPEG. Expected outcome: Reject mismatches; accept supported valid signature. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/unit.xml.

26.2.6 DB01 FR-001 and FR-010

Basis and selection: FR-001 / FR-010. Protect the first persisted request and initial audit record. Technique: State transition.

Precondition and input: Valid new request. Expected outcome: Submitted status, timestamp and initial history saved. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.7 DB02 NFR-001 and NFR-002

Basis and selection: NFR-001 / NFR-002. Prevent cross-requester and administrative data disclosure. Technique: Equivalence partitioning.

Precondition and input: Other requester and platform admin read. Expected outcome: Unauthorised business read denied. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.8 DB03 NFR-002

Basis and selection: NFR-002. Enforce least privilege for sensitive information. Technique: Decision table.

Precondition and input: Scoped roles with different sensitive clearance. Expected outcome: Sensitive access follows category and clearance. Actual result: the expected assertions

passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.9 DB04 FR-010 and NFR-003

Basis and selection: FR-010 / NFR-003. Preserve accountable lifecycle progression. Technique: State transition.

Precondition and input: Accept, assign, progress, resolve and close. Expected outcome: Permitted lifecycle records audit history. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.10 DB05 FR-010 and NFR-005

Basis and selection: FR-010 / NFR-005. Prevent inconsistent state after invalid progression. Technique: State transition.

Precondition and input: Skip a required transition. Expected outcome: Reject and preserve history/version. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.11 DB06 NFR-005

Basis and selection: NFR-005. Avoid silently overwriting newer work. Technique: Equivalence partitioning.

Precondition and input: Stale expected version. Expected outcome: Reject lost update. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.12 DB07 FR-011 and NFR-002

Basis and selection: FR-011 / NFR-002. Protect internal staff information. Technique: Decision table.

Precondition and input: Requester reads private staff comment. Expected outcome: Private comment hidden. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.13 DB08 NFR-001

Basis and selection: NFR-001. Prevent requesters bypassing staff permissions. Technique: Equivalence partitioning.

Precondition and input: Requester attempts staff mutation/direct update. Expected outcome: Mutation denied. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/integration.xml.

26.2.14 E2E01 FR-001 and FR-003 and FR-004

Basis and selection: FR-001 / FR-003 / FR-004. Verify the integrated requester journey and private retrieval. Technique: End-to-end scenario.

Precondition and input: Requester submission, validation, private attachment and another-user read. Expected outcome: Save and retrieve own request; deny other requester. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/e2e.json.

26.2.15 E2E02 FR-009 and FR-010 and FR-011 and FR-012 and FR-015

Basis and selection: FR-009 / FR-010 / FR-011 / FR-012 / FR-015. Verify cross-role workflow integration and visible resolution. Technique: End-to-end state transition.

Precondition and input: Requester to staff to manager journey. Expected outcome: Integrated assignment, progress, resolution, closure and scoped reporting. Actual result: the expected assertions passed on 15 October 2026. Evidence: Milestone_3/Verification_2026-10-15_Final/e2e.json.

26.3 Negative behavior usability and regression

API01 returns 401 for anonymous details; API02 returns 403 for requester work-directory access; API03 returns generic 404 for malformed IDs; API04 rejects cross-origin mutation. API05 returns 409 and a reload message for a stale version, preserves the saved response and permits a valid current-version update. CRDB08 verifies migration 008 preserves existing reasons, permissions and normal workflow. ATT01/02 and CRDB02/04/05/06 retain cleanup, rollback, scope and privacy coverage.

The responsive case exercises a 390-pixel viewport, reduced motion, keyboard navigation, submission and filter recovery. CRE2E01 and hosted browser checks verify reassignment errors, focus, persisted history, mobile width and requester visibility. These support defined tasks rather than universal accessibility/usability certification. Earlier 11/13 October and pre-correction 15 October reports remain history; F15 contains final results and retained findings.

27 Continuous integration and configuration control

The M3 quality gate run 13 passed for application commit 374c21e7c2b3d9ecc01add9ec2b957ca53139a97, including migration 008 and the detail-read correction. The workflow runs on pull requests and pushes to main: installation, lint, secrets/audit, unit/integration tests, simulation build, browser/API tests, performance, recovery and artifact upload. Run 11 for 5a4e4f4 remains historical evidence; dated local measurements remain separate from CI execution. Successful run 13 and artifacts. Historical run 1 failed its browser step and skipped performance; artifact upload still completed. This records a failed gate blocking normal progression. Historical failed run 1

27.1 Review and protection evidence

Protected main and two non-author reviews remain the governance policy. Branch protection is enforced, as confirmed by Ryno. PRs 30-32 are merged with two approvals each. PR34, containing the final fixes, verification and PED, was approved by Steven and Willem and merged on 16 October. These reviews and CI13 support the identified application baseline; they do not certify production fitness.

PR	Contribution and reviewers	CI evidence
30	Ryno architecture/recovery; approved by Steven and Willem; merged	Run 5 passed
31	Steven data/traceability; approved by Ryno and Willem; merged	Run 7 passed
32	Willem quality/release; approved by Ryno and Steven; merged	Run 9 passed
33	CR-M3-01; Steven and Willem approval confirmed by Ryno	Run 11 passed
34	Final corrections and evidence; approved by Steven and Willem; merged 16 October	Run 13 passed

27.2 Static security and dependency interpretation

ESLint, the bounded secret-pattern scan, positive/negative access tests and the strict UI audit passed. The 15 October production audit reports zero vulnerabilities. The 13 October full audit records five high-severity development dependency nodes rooted in braces advisory GHSA-vfj7-8cjw-p6xm. These are dated findings, not five distinct proven application exploits. The then-listed automated remedy was an incompatible older lint configuration, so no forced downgrade was applied. Development-tool exposure remains documented and re-evaluation belongs to dependency maintenance. No penetration-test or complete security certification is claimed.

Threat controls focus on session-derived identity, row-level scope, sensitive clearance, origin checks, transaction integrity, least-privilege writes and private attachment/reason access. Direct reason-column selection is denied; m3_assignment_history applies visibility rules. Recovery integrity uses file hashes and rejects corruption. Hosted privileged-credential configuration remains an operational exposure beyond the local simulation.

28 Performance staging and recovery evidence

28.1 Performance workload and result

The final local authenticated own-list scenario used 1000 requests and three identities. Three concurrent workers each ran 20 sequential reads after warmup: 60 reads, approximately 333 rows per response and zero errors. Its p95 was 72.15 ms against the unchanged 2000 ms read target. Local performance evidence.

Retrieval is performance-sensitive because users must find their work promptly. This exercise supplies the manageable M3 load evidence while preserving the M1 targets. It does not prove all operations at 1000 hosted rows or sustained production capacity. NFR-006 therefore remains partial across its complete acceptance scope; the 2000 ms read and 3000 ms submission targets are unchanged.

28.1.1 Hosted workload and measured outcomes

A production Next.js build connected to hosted Supabase measured 60 list/management reads and 15 detail reads using three concurrent authenticated identities, plus six browser submissions and 45 browser search/filter/sort interactions. At measurement time, requester, second requester and manager could see 9, 3 and 15 records respectively. Filters and sorting operate in the browser after data loads. These small-data measurements are separate from the 1000-row local workload. Values below use nearest-rank p95. Hosted performance evidence.

Operation	Count	p95 ms	Max ms	Target and result
Own request list	40	1232.54	1272.97	2000 ms; passed
Management data	20	1877.22	4185.50	2000 ms; passed
Request details	15	1473.70	1473.70	2000 ms; passed
Browser search	15	45.80	45.80	2000 ms; passed
Browser filter	15	34.10	34.10	2000 ms; passed
Browser sort	15	34.48	34.48	2000 ms; passed
Browser submission	6	1930.99	1930.99	3000 ms; passed

All measured groups met the stated p95 targets with zero operation errors. One management read took 4185.50 ms; 19 of its 20 reads were within two seconds. The workload does not establish the larger dataset or sustained-load acceptance scope. Six new performance fixtures were marked Rejected with an explicit fictional-verification reason; real requests were not removed.

The initial hosted detail p95 was 2415.75 ms. After parent authorisation, the service now fetches independent history, comments, assignment history, reopening records and access flags concurrently. The same checks and data protections remain. Remeasurement at p95 1473.70 ms passed. The initial failure and corrected results are retained; transient conflict timing is recorded as HOSTED-TIMEOUT-01 in section 29.

28.2 Coordinated backup and recovery rehearsal

The final controlled 1000-request rehearsal restored and validated a fresh PGlite instance in 468.60 ms against the 30-minute target. Its simulated loss window was 507 ms against the 24-hour threshold. The coordinated snapshot includes data/history, assignment reasons,

comments, reopening records, identities/scopes, reference data, attachment bytes and non-secret configuration. Local recovery evidence.

The original database was closed and in-memory objects cleared before restore. Exact table comparison, attachment bytes/type, requester isolation, internal comment/reason visibility and continued workflow passed. A post-backup request was correctly absent, demonstrating loss after the snapshot. A corrupted archive was rejected. Versioned files and SHA-256 hashes are published through a manifest only after all snapshot files are written. Command: npm run test:recovery.

This verifies a coordinated local snapshot taken while mutation is paused. Routine daily scheduling, changing-database backup, hosted database/Storage/Auth restore and production recovery guarantees remain unverified. Public application credentials do not provide administrator restore access. These operational shortcomings are scheduled for the final submission; the current rehearsal is not presented as their completion.

28.3 Operational visibility and rollback

Local application/provider output, explicit UI failures and retained test reports provide basic demonstration fault visibility. Hosted operation would require controlled error/access-denial review without logging tokens, credentials or sensitive content. This remains a monitoring concept rather than production alerting. Recovery must coordinate data, objects, configuration and a compatible application release; migration rollback implications are recorded in section 30.

29 Residual risk technical debt and defects

Section 20 is the current risk register, preserving R1 through R16, owners and ratings. Passing tests mitigate selected failures but do not close every risk. The residual concerns below explain the remaining release conditions.

Residual concern	Mitigation and remaining limit	Owner and next review
Hosted parity	Selected hosted Auth, database, private Storage and browser cases passed; full operational equivalence and monitoring remain for final submission.	All; M4 environment review
Incomplete performance scope	Measured local list and small-data hosted operations passed; all-operation 1000-row hosted and sustained-load coverage remain incomplete.	Willem; M4 verification review
Operational recovery	Local coordinated snapshot/restore passed; routine schedule, live backup and hosted recovery remain final-submission work.	Steven and Willem; operational review
Bounded reporting	Current list/report query covers up to 1,000 records; larger sets can truncate summaries.	Steven; data/query review
Development dependencies	Dated high findings retained; production audit passes; no incompatible forced downgrade.	All; dependency review
Network response variability	One post-workload conflict request exceeded 10 s; direct and fresh production-route checks passed. The transient cause remains unconfirmed.	Ryno and Willem; final-submission reliability review

29.1 Technical debt register

ID	Debt and consequence	Owner and review point
TD-M3-01	Selected hosted verification passed; wider parity, production configuration and monitoring remain incomplete.	All; M4 environment validation
TD-M3-02	Local 1000-row list and small-data hosted timing passed; all-operation hosted dataset/load coverage remains incomplete.	Willem; M4 quality review
TD-M3-03	Routine backup scheduling and hosted database/Storage/Auth recovery remain unverified.	Steven and Willem; operational readiness
TD-M3-04	1,000-record query limit may truncate larger reports.	Steven; query/pagination planning
TD-M3-05	Reason-bearing history consumers must use the protected RPC; legacy select-all access is intentionally incompatible.	Steven and Ryno; interface review
TD-M3-06	Transient post-workload timeout needs wider repetition, correlation and network/provider diagnosis.	Ryno and Willem; final-submission reliability review

29.2 Defect and verification finding register

ID and basis	Observed condition	Disposition and evidence
DEF-M3-01; M2 section 18.2	Historical example-route build failure prevented that baseline build.	Resolved in the tested M3 candidate; CI11 build passed. Prior failure remains in the M2 snapshot.
VER-CR01-01; CRE2E01	Initial test alert selector matched both assignment error and Next route announcer.	Alert selector corrected; historical eight-case result and CI11 retained. Final nine-case local browser/API suite also passed.

ID and basis	Observed condition	Disposition and evidence
DEF-M3-02; NFR-005	Stale business versions used SQLSTATE 40001 and could trigger hosted retries/timeouts.	Migration 008 uses PT409; service maps 409 with reload message. DB06, CRDB08, API05 and focused hosted checks passed.
PERF-DETAIL-01; NFR-006	Initial hosted detail p95 2415.75 ms exceeded the 2000 ms target.	Independent authorised reads execute concurrently; remeasured p95 1473.70 ms passed; local regressions passed.
VERIFY-FILTER-01; hosted harness	Initial exact Status label selector failed to locate the nested form select.	Harness targets existing filter controls. All 45 follow-up search/filter/sort interactions completed. Original result retained.
HOSTED-TIMEOUT-01; R9	One post-workload stale request exceeded a 10-second client timeout.	Direct RPC retest returned PT409 in 271 ms; fresh production route returned 409 with unchanged state in 2247 ms. Cause unconfirmed; retained for final submission.

Verification defects are distinguished from application defects and network observations. The original alert-selector failure remains in earlier evidence. Initial hosted measurements and selector failure and Final hosted results and retained timeout preserve the later findings. Passing targeted retests does not prove that transient latency or all defects have been eliminated.

30 Controlled change and impact analysis

CR-M3-01 is the team-selected simulated client change for required reassignment reasons. All three members jointly decided on and approved it on 15 October 2026. The scenario is recorded as simulated; it is not attributed to an externally issued lecturer/client request. PR 33 and candidate commit provide controlled implementation evidence.

30.1 Change control record

Field	Record
Date	15 October 2026
Type	Simulated client change request jointly selected by all three team members
Request source	Ryno Goetz, Steven Riaan Piek and Willem Booysen jointly decided on the reassignment-reason change
Implementation approval	Ryno Goetz, Steven Riaan Piek and Willem Booysen jointly approved implementation on 15 October 2026
External approval	This request is not represented as issued or approved by the lecturer or an external client
Implementation branch	Ryno_branch
Implementation review	Steven Riaan Piek and Willem Booysen approved; confirmation provided by Ryno on 15 October 2026. GitHub evidence: PR 33.
Governing baseline	Master Project Brief section 20.3; PED v2.0 FR-009, FR-016, NFR-001, NFR-002, NFR-003, NFR-005 and NFR-008

30.2 Formal simulated request

When a manager changes a request from one responsible staff member to another, require a reason for the change. Record that reason with the assignment so that authorised users can understand why responsibility changed. First assignment must remain possible without a reason. Preserve existing permissions, internal-note visibility and request history.

30.3 Before and after

Previously, a permitted reassignment could be saved without an explanation. An optional assignment note was stored as a comment, but the assignment record did not contain the reason.

After this change, a different assignee requires a non-whitespace reason of no more than 4,000 characters. Initial assignment and retaining the same assignee keep the note optional. Accepted assignment notes are stored with the new assignment, its actor and timestamp; the existing comment record is retained. Earlier assignments and comments are preserved. Legacy rows have no invented reason.

30.4 Controlled acceptance criteria

FR-009 retains its original five acceptance criteria. CR-M3-01 adds supplemental criterion AC6: changing an existing responsible staff member to a different staff member requires a valid reason, saved with the attributable assignment history. The existing FR-009 permission criteria remain authoritative.

Case	Required outcome
First assignment with no reason	Permitted when the existing role, category, sensitivity and lifecycle rules allow it
Save the same assignee	Reason remains optional
Different assignee with an omitted, empty or whitespace-only reason	Rejected before mutation; assignment, target, lifecycle and version remain unchanged
Valid reason	Trimmed and saved with the new assignee, acting user and assignment timestamp
Length boundary	4,000 ASCII characters accepted; 4,001 rejected. Browser and service validation also bound JavaScript input length
Existing assignment history	Retained without rewriting prior reasons or inventing missing ones
Unauthorised, out-of-scope or stale operation	Rejected even when a valid reason is supplied
Internal reason	Available only to authorised operational users; not disclosed through requester history or direct column reads
Requester-visible reason	Visible through the authorised request detail and assignment history

30.5 Impact analysis

Area	Impact and decision
Business need	Extends accountability for changes in responsibility under NEED-005 and NEED-008
Requirements	FR-009 receives AC6; FR-016 and NFR-003 gain reason-history evidence. Stable requirement IDs and original acceptance criteria are retained
Interface	The existing Assignment form identifies required reassignment reasons, shows inline errors, focuses the reason field and preserves input after failure. Assignment history displays saved reasons with actors and timestamps
Application boundary	Shared assignmentReasonError supports browser and server checks. The database independently enforces the rule so direct RPC callers cannot bypass it
Persistence	Additive migration 007 adds reason and reason_requester_visible to assignments, updates the existing atomic workflow function and adds m3_assignment_history
Privacy and access	Existing manager/staff powers, category scopes and sensitive clearance are retained. Direct access to the new reason columns is restricted. The history RPC returns internal reasons only to callers allowed to work on the request
Integration	The request-detail service reads assignment history through the protected RPC. The simulation provider implements the same RPC contract
Verification	Unit boundary cases, database policy/transaction cases, migration-upgrade testing, a browser/API journey and recovery regression cover the change

Area	Impact and decision
Architecture and decisions	The modular monolith, ADR-001, ADR-002 and central lifecycle policy remain. D10 gains the assignment-history contract; D12 gains migration 007. A separate architecture ADR is unnecessary for this bounded extension
Dependencies and cost	No package or paid service is added. Change cost is limited to the existing form, service, migration, simulator and verification records; no unmeasured effort estimate is claimed
Delivery	CR-M3-01 originally required migration 007 before its changed service. The final application also requires the migration 008 version-conflict correction. Fresh setup applies 001 through 008 in order.

30.6 Implementation evidence

Responsibility	Repository evidence
Shared validation	src/domain/requests/assignment-reason.js
Server validation and protected history	src/features/workflow/service.js
Form, inline feedback and history	src/features/workflow/request-workspace.js
Persistent enforcement and privacy	supabase/migrations/007_m3_reassignment_reason.sql
Local provider contract	tests/support/provider.mjs
Migration-aware fixtures	tests/support/database.js

30.7 Requirement and verification trace

Requirement or concern	Test evidence	Expected result
FR-009 AC6 and first-assignment compatibility	CR01, CR02, CR03; CRDB01, CRDB02, CRDB04	Optional first note; required valid reassignment reason; clear boundaries
FR-016 and NFR-003 attributable history	CRDB03; CRE2E01	Original assignment retained; new assignee, reason, actor and timestamp retrievable after reload
NFR-001 permissions and NFR-005 stale updates	CRDB05	A reason does not bypass authority, scope or expected version
NFR-002 internal information	CRDB06	Internal reason hidden from requesters and protected against direct column reads
NFR-005 atomicity	CRDB02, CRDB04, CRDB05	Failed operations preserve request and assignment state
NFR-008 controlled migration	CRDB07	Existing assignment data and comments survive the upgrade; new rule then applies
NFR-004 form feedback	CRE2E01	Required-field error is visible and associated; focus moves to the reason; mobile layout remains usable
NFR-007 recovery regression	scripts/test-recovery.mjs	Reassignment reason survives snapshot/restore with its visibility restrictions

Unit cases are in tests/assignment-reason.test.js, database cases in tests/integration/reassignment.test.js, and browser/API cases in tests/e2e/workflow.spec.js. F15 indexes final execution. API05 and CRDB08 extend integrity and migration coverage after CR-M3-01; they do not change FR-009 AC6. Earlier 13 October and initial 15 October reports remain historical evidence.

30.8 Risk and debt continuation

R11 remains open: request, assignee, actor, reason and version must stay consistent. Transactions and failure tests mitigate it. R12 retains the stale-update risk; migration 008 and hosted conflict checks verify the corrective response. R3/R14 retain disclosure risks with selected hosted access and reason-visibility evidence. Wider hosted assurance, routine backup and recovery remain scheduled for the final submission.

The new RPC adds a small compatibility obligation. Consumers must use m3_assignment_history for reason-bearing history; direct select of every assignments column is intentionally denied to authenticated clients. Public legacy metadata columns retain column-level read permission under the existing row policy. No existing risk is automatically closed and no acceptance criterion is waived.

30.9 Deployment and rollback

The simulation applies the migration files automatically. The hosted test operator applied migrations through 008; direct hosted conflict checks confirmed PT409. Fresh hosted setup requires migrations 001 through 008 in order. Application build settings must match the selected provider. An earlier applied migration is not silently rewritten to deploy this correction.

For a controlled rollback, pause writes and preserve coordinated data, attachments, configuration and the matching application release. Preserve post-backup history where possible. Do not deploy the old detail reader against reason-column protections, and do not reintroduce retryable 40001 for a business version conflict. A forward correction is preferable where it preserves history and privacy. No destructive rollback or hosted disaster-recovery exercise was performed.

31 AI usage contributions and approvals

The M1 and M2 AI registers in sections 12 and 23 remain dated history. The following M3 continuation identifies actual assistance and the verification applied. Tool output remains subject to team accountability; a passing check does not replace engineering judgement.

Date and responsibility	Assistance	Verification and outcome
11 October; Willem	Codex integration of attachment/reporting and quality checks; commits 91ca393 and c055bdf.	21 unit and 14 DB cases passed on that branch; browser/performance then blocked by loopback EACCES. Later integrated execution superseded that blocker.
11-13 October; Ryno	Codex frontend integration, approved visual redesign, recovery verification and shared PED contribution drafting.	Integrated 42-case local suite, build, lint and security checks passed. PR30 approved by both other members; contribution documents reviewed through PR30-32.
13 October; team contributions	Shared AI-assisted drafting produced the three member documents; authors reviewed their contribution areas.	PR30-32 record the reviewers and approvals. Each contributor remains accountable for the accepted content.
15 October; Ryno for team CR-M3-01	Codex inspected workflow/privacy, implemented reason validation/history and migration, tests and impact record.	53 local tests, recovery/performance/build/static checks and CI11 passed. Alert-selector ambiguity corrected with failure evidence retained. Joint change approval confirmed by all three through Ryno.
15-16 October; shared PED integration	Codex integrated PED v3.0, updated README setup guidance and reconciled source, review and CI references.	Compared with source briefs, controlled contributions, test reports and GitHub records; rendered document inspected for layout and text integrity.
15 October; Ryno verification and corrections	Codex tested hosted accounts, policies, attachments, workflow and performance; implemented PT409 mapping/migration and concurrent detail reads.	55 final local cases and focused hosted conflict checks passed. Detail p95 improved to 1473.70 ms; original failures and transient timeout retained. Ryno scheduled remaining shortcomings for final submission.

31.1 Contribution and decision approval evidence

Member	M3 contribution	Approval evidence
Ryno Goetz	Architecture, frontend, staging, recovery and controlled reassignment implementation.	PR30: Steven and Willem approval; CR-M3-01 jointly approved 15 October.
Steven Riaan Piek	Data/persistence, backend workflow and evolved requirements trace.	PR31: Ryno and Willem approval.
Willem Booysen	Attachment/reporting rules, quality checks, CI and release evidence.	PR32: Ryno and Steven approval.

Ryno confirmed PED approval and enforced branch protection on 13 October. All three members jointly approved the simulated change on 15 October; Ryno confirmed both non-author implementation approvals. PR34 records Steven and Willem approving the final fixes and PED before merge on 16 October. These approvals concern the engineering baseline, not lecturer acceptance or production fitness.

32 Release evidence summary and submission baseline

Classification: CONDITIONALLY READY for the selected M3 demonstration. The core workflows passed local and selected hosted checks; final local regression totals 55 cases. Local 1000-row performance/recovery and small-data hosted timing passed. CI13 passed for the final application commit and PR34 was approved by both non-author members and merged. Remaining operational and broader acceptance limits are scheduled for final submission.

Evidence area	Result and interpretation	Residual concern
Build and CI	Simulation/hosted builds and local quality gates passed. CI13 passed for 374c21e; PR34 approved and merged.	CI supports its configured checks, not complete acceptance or production readiness.
Automated regression	24 unit + 22 DB + 9 browser/API = 55 local cases passed.	Selected coverage; complete criterion acceptance is not asserted.
API and system	Five local API cases, three journeys and responsive case; ten selected hosted browser checks and focused conflict results.	Hosted testing covers fictional scenarios, not full deployment certification.
Static and security	Lint/secret/UI checks pass; current production audit has zero findings.	Dated development findings and hosted credential/policy operation remain exposure.
Performance	Local list p95 72.15 ms at 1000 rows/3 identities. Hosted p95 list 1232.54, management 1877.22, details 1473.70 and submission 1930.99 ms.	Hosted measurements used small datasets; larger workloads and variability remain final-submission work.
Recovery	1000-request coordinated local restore 468.60 ms; simulated loss 507 ms.	Routine scheduling, live backup and hosted recovery remain unverified.
Change and acceptance	CR-M3-01 approved; reason rule, history, privacy and migration tested.	Retention and other stated partial criteria remain explicit in RTM.
Defects and debt	Version-conflict retry and slow detail retrieval corrected; harness findings and transient timeout retained.	Timeout cause unconfirmed; residual risk and technical debt remain explicit.

32.1 Exact release identification and access

Repository: https://github.com/RynoGz/SEN831_Group_F. Application version: 0.3.0. Verified application commit: 374c21e7c2b3d9ecc01add9ec2b957ca53139a97. PR34 merged this candidate from Ryno_branch into main as 9e71bd9f230328ef4a0268c64402b370c9ae20ea. CI13 passed for the application commit. The dated F15 manifest records the original collection base 4c35ffc and the source/report hashes; all recorded source hashes match the committed application. Final verification manifest. Manifest SHA-256: bb8c0b717a53d0b211a0c1805078512d47ae93076cb23a4cf823e2d9a0cc4b87. Documentation-only revisions do not change this application identifier. Local staging is in section 24.4 and hosted setup in section 24.6.

32.2 Evidence index

Evidence	Controlled location
Candidate CI and artifacts	Successful run 13 with published verification artifact
Source and execution hashes	Final source and evidence hash manifest

Evidence	Controlled location
Unit and database	24 unit results; 22 database results
Browser and API	9 local browser/API results; 10 hosted browser checks; Focused hosted conflict result
Performance and recovery	Local performance; Hosted performance; Local recovery
Security and UI audit	Production audit; UI contract audit
Controlled change	CR-M3-01 request and impact analysis
Review trail	PR30; PR31; PR32; PR33; PR34
RTM and registers	Sections 19-22, 29 and 31 in this continued PED; historical snapshots retained earlier.
Initial finding evidence	Initial hosted timings and selector failure; final hosted report retains post-workload timeout and successful retest.

The PED v3.0 PDF is the primary milestone document and evidence index. The final verification manifest identifies the tested files independently of historical CI and later documentation changes. The remaining operational, load, retention, monitoring and reliability work is recorded for the final submission with owners in section 29. No incomplete criterion or unexecuted recovery scenario is represented as passed.

References

S1
SEN381 Teaching Team. SEN381 CivicConnect Project Milestone 1 Engineering Foundation and Requirements Baseline. 2026.

S2
RynoGz SEN831 Group F repository at commit 1a13a01, including merged PRs 22 to 25, reviewed 30 September 2026.

S3
Team confirmations recorded by Ryno Goetz covering membership, working arrangements and approval of the M2 decisions.

S4
SEN381 Teaching Team. CivicConnect Master Project Brief Community Service Request Management Platform. Version 1.1. 2026.

S5
ISO IEC IEEE 29148 2018. Systems and software engineering Life cycle processes Requirements engineering.

S6
ISO IEC 25010 2023. Systems and software engineering Systems and software Quality Requirements and Evaluation Product quality model.

S7
SEN381 Teaching Team. CivicConnect Milestone 2 Architecture Technology Initial Design and Application Foundation. 2026.

S8
Milestone 2 Member 1 CivicConnect M2 Ryno Architecture and Frontend version 0.2 and frontend handoff. 30 September 2026.

S9
Milestone 2 Member 2 ADR-001 Technology Stack Decision. Accepted 30 September 2026.

S10
Milestone 2 Member 2 M1 Requirements Review and Data Model Baseline. 30 September 2026.

S11
Milestone 2 Member 2 ADR-002 Database and Persistence Decision and migrations 001 to 004. 30 September 2026.

S12
Milestone 2 Member 3 Design and Register Updates and request transition policy. 30 September 2026.

S13
Milestone 2 Member 2 CivicConnect M2 Requirements Traceability Matrix version 0.4. 30 September 2026.

S14
CivicConnect Assignment 2 and Assignment 3 consolidated research artefacts. September 2026.

S15
Next.js documentation. Authentication and App Router guidance. nextjs.org docs.

S16
Supabase documentation. PostgreSQL database, Row Level Security, database functions, Storage access control and backups. supabase.com docs.

S17
Vercel documentation. Deployment environments. vercel.com docs.

S18
SEN381 Teaching Team. CivicConnect Milestone 3 Controlled Construction Integration Quality and Release Readiness. 2026.

S19
Approved M3 member contributions in Milestone_3; PRs 30, 31 and 32, 13 October 2026.

S20
CivicConnect application commit 374c21e7c2b3d9ecc01add9ec2b957ca53139a97 and successful GitHub quality gate run 13. PR34 records both non-author approvals and merge on 16 October 2026. Run 11 for earlier CR-M3-01 candidate 5a4e4f4376908c6750ce9dd7db20674feb78a843 remains historical evidence. Run 13; PR34

S21
CR-M3-01 Reassignment Reason, jointly selected and approved 15 October 2026; controlled request record

S22
Local verification reports dated 15 October 2026; controlled evidence index

S23
Willem M3 AI usage register, 11 October 2026; subsequent integration and verification history in this PED and PRs 30-33.

S24
Final local and hosted verification, 15 October 2026. Controlled verification manifest. Local 55-case suite, hosted functional/conflict checks, performance measurements and local recovery evidence.

S25
Supabase documentation. SQLSTATE 40001 in an RPC function causes retries; custom PT409 returns a business conflict without serialization retries. https://supabase.com/docs/guides/troubleshooting/high-cpu-and-infinite-transaction-retries-when-using-custom-error-codes-in-rpc-functions-77326b